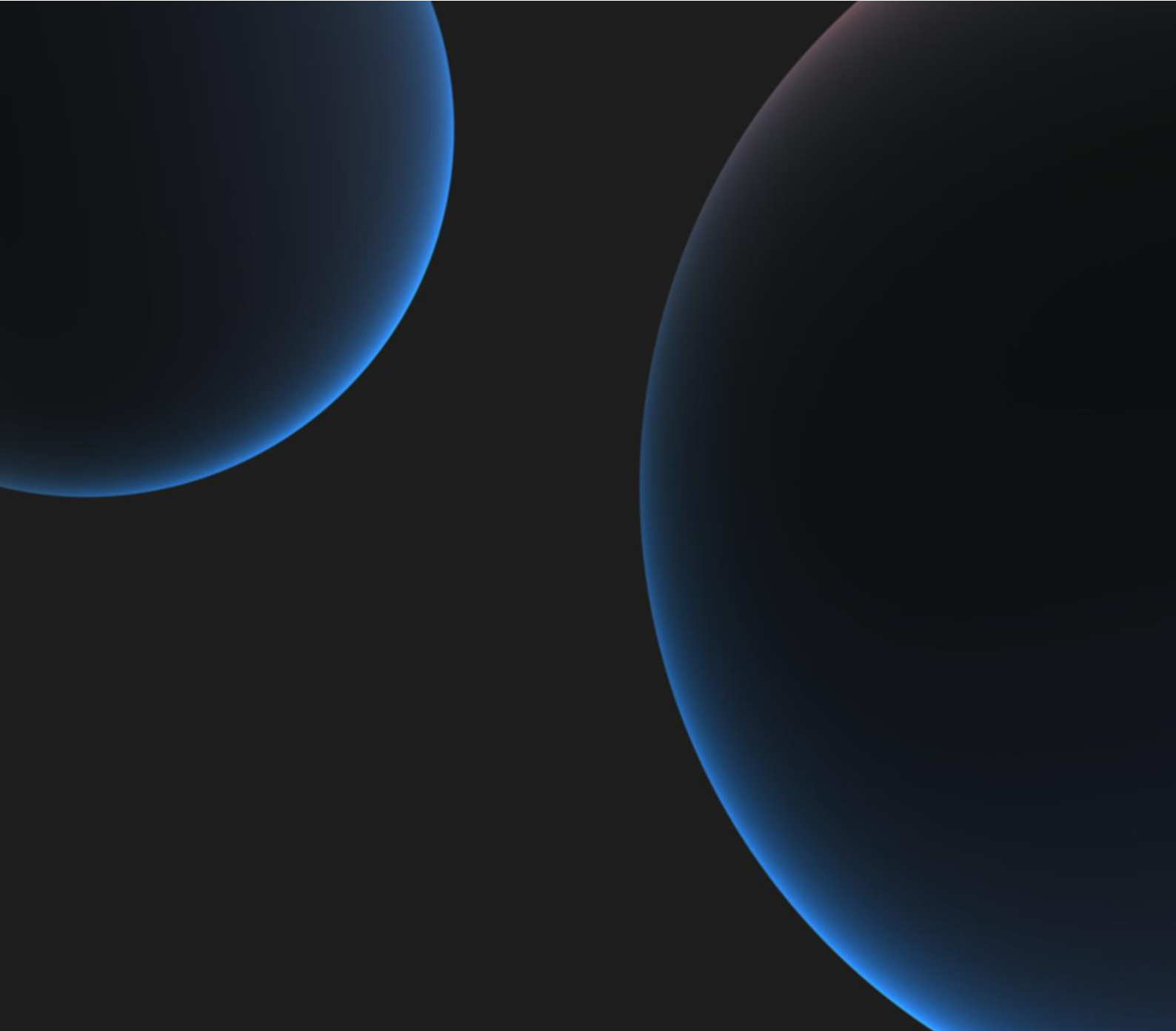




EnOS™ BESS SCADA

Product Manual

v2.4

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Foreword

About this document

This document is used to introduce the main functions, system performance, and software & hardware environment required for system operation of the EnOS™ BESS SCADA system. It is helpful for users to quickly understand the main functions and principles of the product for decision making; it is helpful for solution personnel to evaluate and formulate appropriate solutions; it is also helpful for project implementation personnel to reference for the acceptance testing of product implementation results.

Disclaimer

The system graphic screens given in this manual are for reference only, which are subject to project implementation and delivery. Due to continuous product improvement and upgrade, the description of product functions in this manual is subject to change without notice.

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1 Overview

The EnOS™ BESS SCADA system, developed by Univers, is grounded in the principles of smart power, energy interconnection, and sustainable development. It operates on the integrated EnOS™ SCADA platform, which overcomes the technical limitations of traditional SCADA systems by spanning multiple domains, including wind power, solar energy, and energy storage. This enables comprehensive power generation monitoring and energy management to achieve multi-energy data access and interconnection within a unified platform.

Designed with second-level monitoring technology and power generation site models, the system fully accesses data from energy storage devices. It facilitates operations management and power generation control in smart sites through centralized monitoring, intelligent alarms, and advanced data analysis and statistics.

For uplink communication, the system can collaborate with the power grid to relay important device data as needed. For downlink communication, it interfaces with the local controller of the energy storage subsystem to access various device data. This functionality supports panoramic site monitoring, data analysis, intelligent alarms, and remote start/stop control, enhancing the automatic safety management of energy storage sites. It enables a no-duty or minimal-duty management mode, effectively reducing the operation and maintenance costs of power stations.

1.1 Terminology

Abbreviation	Full Name	Description
AGC	Automatic Generation Control	Refers to the automatic power generation control system
AVC	Automatic Voltage Control	Refers to the automatic voltage control system
BMS	Battery Management System	Used for monitoring battery voltage, current, temperature and other parameters as well as managing and controlling the battery status
ECC	Energy Collaborative Control	Energy Collaborative control module
EMS	Energy Management System	Refers to an energy management and dispatching system provided by Univers for active and reactive power control of wind power, solar and energy storage.
ESS	Energy Storage System	Refers to a set of minimum energy storage subsystem that consists of a single PCS and corresponding battery pack in the energy storage power station

LCRS	Local Comprehensive Recording System	Refers to a system designed for the thorough recording of data or events within a specific local area.
PCC	Point of Common Coupling	Refers to the public connection point between the booster station in the power station and the power grid
PCS	Power Conversion System	Used in electrochemical energy storage system to connect the battery and the grid to achieve bidirectional electrical energy conversion
PFC	Primary Frequency Control	Refers to a system for Primary Frequency Control
PMU	Phasor Measurement Unit	Refers to the synchronous phasor measurement unit
PPC	Power Plant Controller	Power plant control for the whole station
SCADA	Supervisory Control and Data Acquisition	Refers to the SCADA platform, which is a comprehensive monitoring application software product for business scenarios such as wind farms and energy storage developed by Univers.
SOE	Sequence of Event	Millisecond-level sequential records of events that occurs at each measurement point (column) within the selected time range, which are the most important basis for device/system failure and abnormal analysis
TBA	Time Based Availability	Time-based availability.

1.2 Target users

This document is prepared mainly for the following readers for easing their related business.

- Solutions
- Sales team
- Delivery team
- Technical personnel for end customers

1.3 Application scenarios

The EnOS™ BESS SCADA system is mainly used for monitoring the devices at the site control layer and the spacer layer.

- By monitoring and controlling the devices in the energy storage site, including energy storage subsystems, energy storage converters and energy storage batteries, it can achieve the monitoring and management of the assets in the whole site.

- The managers can understand the real-time and historical operating data of the wind farm at any time through the PC devices.
- The technicians can analyze alarms, power curves and other data through the system.
- The maintenance personnel can use the remote diagnosis service provided by the system for rapid response to device faults so as to timely monitor and online analyze faults and reduce downtime.
- The operators can conveniently view the operations of all the devices across the site through the monitoring system.

1.4 System features

The EnOS™ BESS SCADA system is designed in accordance with key international and regional standards to ensure compliance with communication protocols, power system interoperability, cybersecurity.

It supports industry protocols such as IEC 60870-5-104, ModbusTCP, IEC61850, OPC-UA for communication. To ensure system security, it adopts best practices aligned with IEC 62443 for industrial cybersecurity and ISO/IEC 27001 for information security management.

Its development process follows a structured approach based on CMMI-DEV Level 3, ensuring consistent quality, traceability, and continuous improvement.

These standards help ensure smooth integration with third-party systems and compliance with utility and regulatory requirements across global markets.

Its main features include:

- By monitoring and controlling the energy storage subsystems, box transformers, converters, battery arrays, and cells in the energy storage site, it can achieve the intelligent monitoring and management of the assets in the whole site.
- The managers can understand the real-time and historical operating data of the energy storage site at any time through the PC terminal.
- The technicians can analyze device alarms, power curves and other data and failures through the system.
- The maintenance personnel can use the remote diagnosis service provided by the system for rapid response to device faults so as to timely monitor and online analyze faults and reduce downtime.

- The operators can conveniently view the operations of all the devices across the site through the monitoring system.
- Highly configurable user interface. The monitoring interface can be customized according to user needs to meet the monitoring requirements in different occasions and levels.
- Reports can be customized to allow users to obtain the information they care about in a timely manner.
- Wide support for OPC, DNP, MODBUS, IEC-104, IEC-101 and other communication protocols.
- Provides real-time curve tool and historical data archive.
- The complete user permission management function helps to distinguish and perform the permission management for the alarm view, remote reset, start/stop machine and operating parameters of the energy storage site.
- It can provide a system interface based on the B/S architecture, and end users can access it through the application screen or browser.

2 Product Overview

The EnOS™ BESS SCADA system is deployed at the site side of the energy storage site. It is mainly used to display the important statuses of energy storage devices, and provides various advanced application functions, such as panoramic automatic monitoring, data analysis, and intelligent alarms of charging and discharging devices. For the uplink, it can cooperate with the power grid to forward important device data or open data interfaces for third parties on demand; for the downlink, it can communicate with the local controller of the energy storage subsystem to access various device data of the energy storage subsystem and provide a data foundation for advanced functions at the site side. The system is featured by stable overall operation, high scalability and compatibility.

2.1 Hardware architecture

The system is designed with an open hierarchical distribution structure, and the devices at the site control layer include application servers and workstations. As the center of data collection, processing, storage and network management at the site control layer, the application server works at the active/standby dual-network mode to ensure high availability. The workstation is the main human-machine interface of the monitoring system, used for displaying graphs and reports, recording events, displaying and querying alarm statuses, querying device statuses and parameters, and issuing control command issuance. The device layer includes the measurement & control device, the battery management system, the auxiliary control system, and the local controller. The protection and measuring-control device provides state quantity acquisition, AC sampling and measurement, and circuit breaker control. The battery management system is used to monitor the statuses of the battery (including temperature, voltage, current, and state of charge) to provide battery management and communication interfaces. The auxiliary control system provides the data of air conditioning, fire protection, and access control to realize the auxiliary protection of the system. The local controller is used for the local data acquisition, storage and control of the energy storage subsystem, and is equipped with voltage, SOC and other protection functions. The typical architecture and specific device distribution of the EnOS™ BESS SCADA system are shown in the figure below.

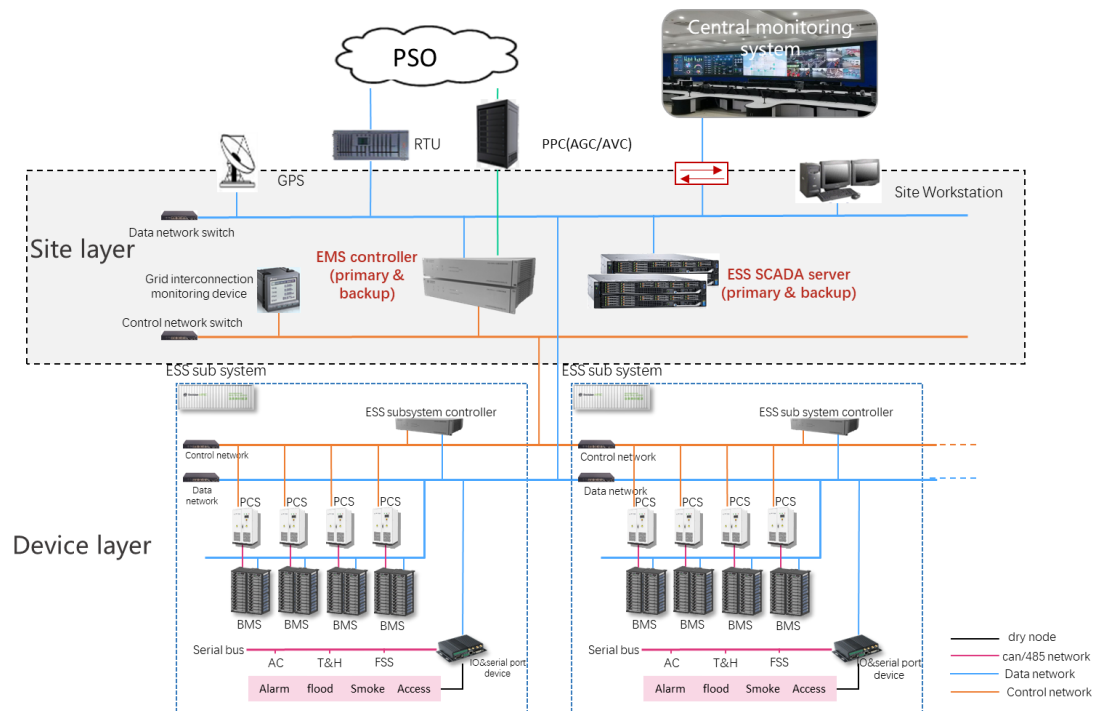


Figure 1: Typical architecture diagram of SCADA system

2.2 Functional framework

The EnOS™ BESS SCADA system is mainly composed of data collection module, data bus, data storage module and advanced application module. The system collects and parses communication data messages, and then stores the device data to the database module and some SCADA advanced applications through the data bus. The applications, including monitoring modules, intelligent alarms and parameter settings, read data from the data storage module for analysis.

- **Data collection:** It supports the connection to multiple types of devices, and forwards data to multiple third-party systems with multiple protocols. The forwarded data supports flexible configuration.
- **Data bus:** By using the message bus mode, it provides data transfer queues for front-end systems, databases and applications.
- **Data storage:** It supports multiple types of data storage methods such as files, memory and databases, and can achieve the category-based data storage in multiple precisions.
- **Monitoring module:** With this function, users can view the site-level and device-level local monitoring information, obtain the real-time operating status of all the devices in the monitored site as well as the parameters of each measurement point, and send remote control commands to control the devices.

- Smart alarm: It is used to confirm, filter and mute real-time alarm information, and it supports the custom configuration of alarm templates, the double pop-up windows for fault alarm and the query and export of historical alarm information.
- Parameter setting: It is used to set various parameters and supports remote Web page setting.

The overall system adopts a three-tier structure designed by standard software that covers the data collection and communication, the data processing and storage, the interface display and operations.

2.3 Function overview

The SCADA energy storage monitoring system is mainly developed for the business areas such as intelligent monitoring and energy management, and provides advanced applications around the site-side and local energy storage subsystems. The intelligent monitoring is intended to solve the needs of customers in the power generation device monitoring and automation control, the alarm monitoring and historical event query, and the device data statistics and core data comparison and analysis.

The L1 functions of the system mainly include monitoring, alarming, reporting and operating parameter tuning, and its detailed L2 functions and descriptions are shown in the table below.

Business domain	L1 functions	L2 functions	Description
Intelligent monitoring	Monitoring	Homepage	Overall power station operation: • Equipment status statistics and links to the equipment list. • Real-time station information. • Operational trends. Subsystem geographical topology: • Drill down to detailed subsystem topologies. • Top-down monitoring and fault location based on subsystem topology.

Business domain	L1 functions	L2 functions	Description
		Subsystem electrical topology	Used to display the composition and connection relationships between feeders, box transformers, and subsystems, as well as the status of switches and disconnectors, subsystem operational status, and key operational data.
		Device list	Used to display a list of various equipment, operational status, and key operational information. Supports card, small icon, and table formats with customizable view order, as well as filtering, tagging, and drilling down to individual equipment details. It mainly includes subsystems, converters, battery packs, battery clusters, air conditioning, monitoring and control, liquid cooling, and other equipment.
		Token list	Used to display all records of tagged monitoring objects based on the equipment list or individual equipment details within the station, with the option to remove tags.
	Alarm	Real-time alarm	Used for the real-time display of system operation information and alarm information.
		Historical alarm	Used to search various alarm information that has occurred.
	Report	Energy Report	Support daily, monthly, and yearly charge and discharge statistics, charge/discharge ratio statistics, and report export for energy storage stations.
		Customized Report	Support custom report templates. Default templates include subsystem hourly operation reports and battery cluster highest cell voltage/temperature reports.

Business domain	L1 functions	L2 functions	Description
	Analytics	Trend analysis	Reflect the changing trends of AI, DI, and PI in the form of tables and curves. - Support custom queries and save templates based on device type and model, such as battery pack voltage difference/temperature difference/current analysis, and battery cluster cell voltage difference/temperature difference analysis. - Support maximum/minimum/average statistics of the queried data. - Support data comparison by day/week/month/year.
		Scatter analysis	Reflects the change trend of the telemetry value in the form of tables and curves.
		Data analytics	Support custom export templates based on device type and model. For example, support exporting battery cluster temperature data, battery cluster voltage data, and battery pack current data.
Settings	Modify password		Modify the password of the current account.
	User management		Covers the user management and role management.
	Alert template configuration		Customize the real-time alarm templates and set the pop-up alarm mode.
	System configuration		Corresponding to the permissions and roles in user management, it is used to configure the device operation and data browsing permissions of different roles.

3 Function introduction

3.1 Monitoring

The monitoring function mainly refers to centralized monitoring and control of energy storage sites. The EnOS™ BESS SCADA system provides the different types of sites with the monitoring pages that support different business needs. See the figure below for the overall monitoring device-level topology of the energy storage system. At the site level, the site homepage is provided for overall overview monitoring; at the subsystem level, the electrical topology of subsystem is provided to monitor the composition and connection relationships among devices; at the device level, the system provides both the list-based and detailed monitoring information of subsystems, converters, battery arrays, battery clusters, air conditioners, and measurement and control devices.

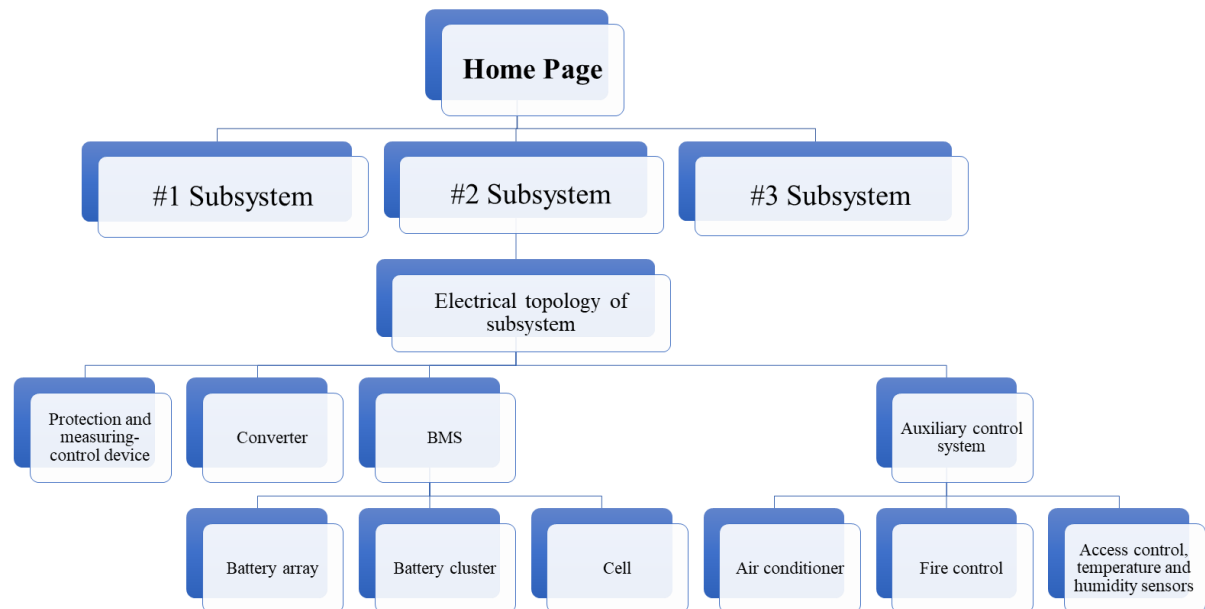


Figure 2: Energy storage monitoring device-level topology

3.1.1 Homepage

The homepage of energy storage site displays the real-time operating information of the site in the form of figures and curves. It provides a unified monitoring platform, which is convenient for users to understand the operation status of energy storage site in time and improve the site monitoring efficiency. The homepage of the energy storage site is shown in Figure 3.

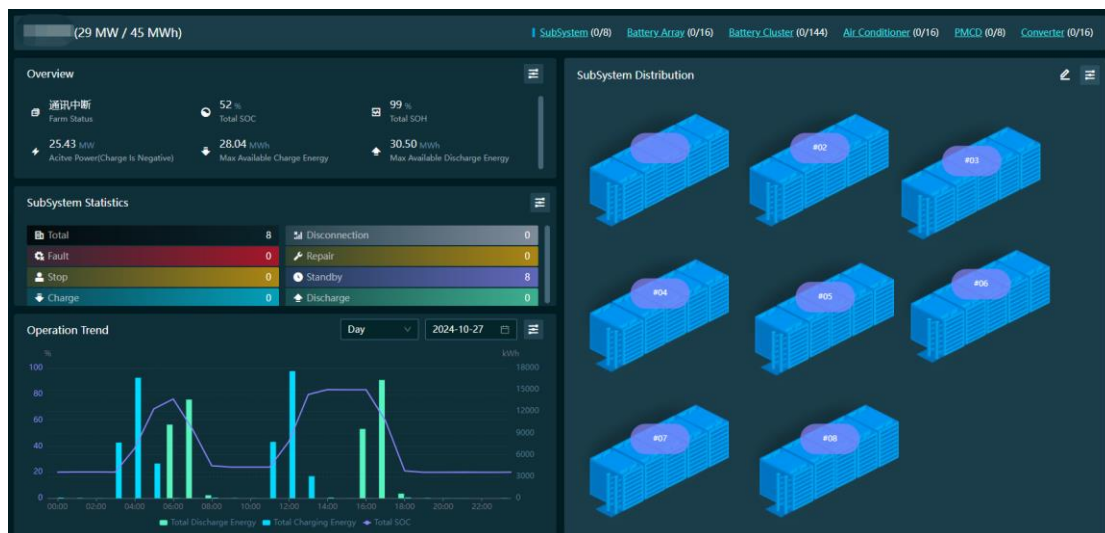


Figure 3: Homepage of energy storage site

The main functions are as follows:

- Station information overview, including the installed capacity of the station, the proportion of faults of various types of equipment, and supporting to click on the device name to drill down to the equipment list for fault screening and positioning.
- Real-time data overview, which provides an overview of real-time data to help users intuitively understand the overall operation of the site.
- Status monitoring statistics, which counts the number of subsystems in different states, and you can directly click on each status statistics to drill down to the device list to view details.
- Curve-based display, which provides the trend curves of important measurement points of the site under different time granularity; the time granularity is customizable and the measurement points can be selected within a certain range.
- Subsystem distribution, which shows the relative distribution positions of all subsystems under the site; you can hover the mouse over important measurement points and status of each subsystem for monitoring, and click a "Subsystem Number" to jump to the electrical topology page of the energy storage subsystem. Click Edit to enter the editing mode, where you can drag and drop the distribution position of each subsystem and edit the display number of the subsystem.

3.1.2 Subsystem of electrical topology

The electrical topology of energy storage subsystem is used to display the connection relationship between the feeder, box transformer and subsystem, the state of the switch, the operation status of the subsystem and the key operation data.

The main functions are as follows:

- Subsystem information overview, including real-time operation status of subsystems and statistics of devices under the subsystem. When the number of faults is greater than 0, it will be displayed in red, and click the corresponding device name to drill down to the device list for fault filtering and locating;
- Electrical topology of subsystem, which is used to monitor the connection relationships among devices, the switch status, and the device operation status. With the topology, users can intuitively view the connection relationships among devices and judge the switch status and the device operation status through the color (the connection interruption is preferred, and the operation status is displayed if the communication is normal). You can click the device to drill down to the device details for each device to locate the problem.
- Device information monitoring, including the key operation information of important devices, such as protection and measuring-control devices, converters, and battery clusters in the subsystem.

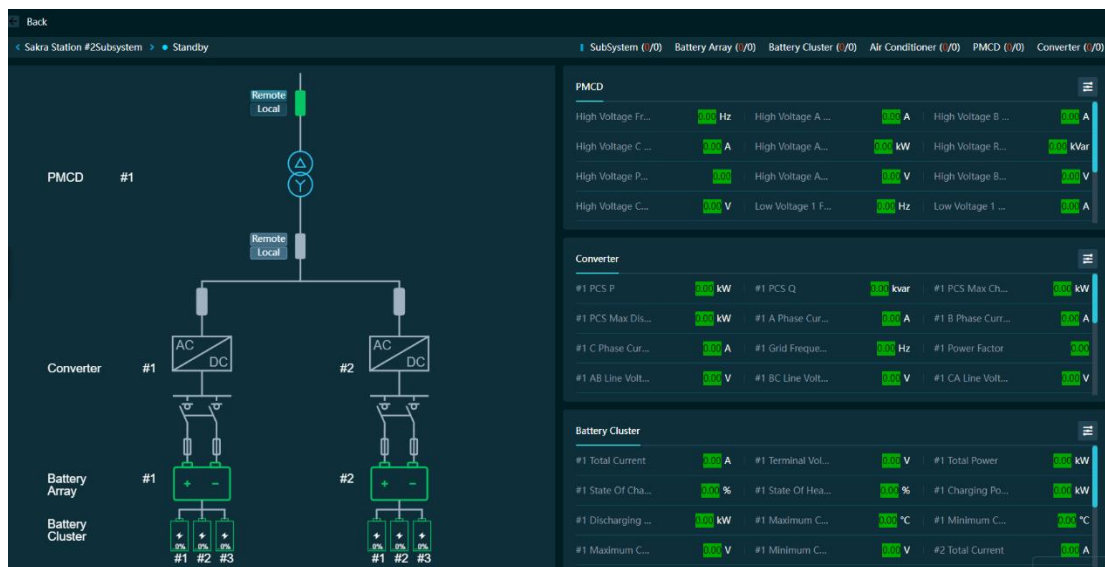


Figure 4: Electrical topology of energy storage subsystem

3.1.3 Electrical topology

The electrical topology of the station displays the connection status of electrical equipment, the state of the switch and the status of the subsystem in the form of wiring diagrams, and provides the monitoring of important measurement points of the box transformers and the subsystems. It is convenient for users to monitor and control the important measurement points of the switch and box transformer of the station, and the electrical topology is shown in the following figure:



Figure 5: Electrical topology of energy storage station

The main functions are as follows:

- The wiring diagram of the station, including the connection status of each box transformer and related electrical equipment, etc.
- Real-time data overview, providing monitoring of all box transformers and important measurement points of subsystems of the power station, helping users intuitively understand the overall operation of the station.
- Support switch name modification to meet the needs of different users for switch definition.
- Measurement point configuration: Support the configuration of different monitoring measurement points and customized display styles to meet the monitoring needs of different users.

3.1.4 Device list

The device list page can be switched through the tabs to display the device lists of different access devices. This page displays the monitoring information of various devices under the site through three ways: large card, small card and list, and you can switch among them through 3 different icons in the upper right corner of different pages. Each mode covers the device status statistics. Also, you can click the configuration button  in the upper right corner of the page for filter configuration, card configuration, and table configuration. The display content and styles can be flexibly adjusted, changed and saved to adapt to the diversity of screen needs. The right side of the filter bar displays the number statistics of each listed object by default and supports filtering, and you can click the name of each device to perform the listing operation.

With the large card-based mode, you can intuitively understand the operating status and key operating indicators of devices, quickly locate abnormal devices, and check the overview of all the operation and statistical information of devices. You can adjust the information to be displayed through the "Card Configuration".

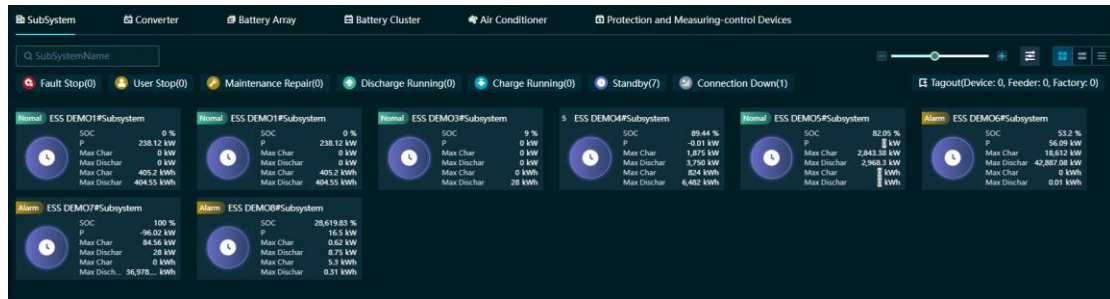


Figure 6: Large card-based device list

The small card-based mode is suitable for scenarios with a large number of devices, such as battery clusters, so as to provide an overview of as many devices as possible and facilitate the timely detection of abnormal operations. The icon color and style are consistent with those configured in the large card-based mode.

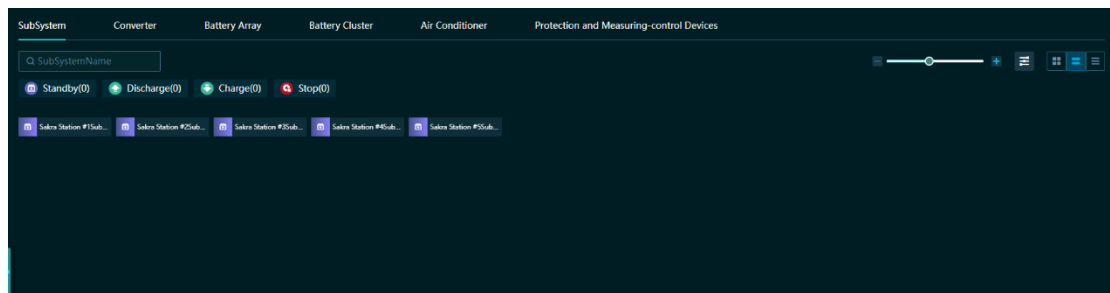


Figure 7: Small card-based device list

The list-based mode supports the display of all devices and their key operating indicators. You can freely sort the status details, active power, reactive power, health, and communication status. You can adjust the information to be displayed through the "table Configuration". Also, it supports setting of benchmarking devices, and benchmarking devices are always on top.

Name	Standard Working Stat...	Health Status	Communication	SOC (%)	P (kW)	Q (kVar)	Max Char (kW)	Max Dischar (kW)
Sakra Station #1Subsy...	Standby	Normal	Break	0.00	0.00	0.00	0.00	0.00
Sakra Station #2Subsy...	Standby	Normal	Break	0.00	0.00	0.00	0.00	0.00
Sakra Station #3Subsy...	Standby	Normal	Break	0.00	0.00	0.00	0.00	0.00
Sakra Station #4Subsy...	Standby	Normal	Break	0.00	0.00	0.00	0.00	0.00
Sakra Station #5Subsy...	Standby	Normal	Break	0.00	0.00	0.00	0.00	0.00

Figure 8: List-based device list

The main functions are described as follows:

- Displays the types of devices connected to the site; you can click each tab to view the list of different types of devices.
- Provides the quick overview of device status, where the devices of different states will be marked with different colors and filter labels for you to quickly filter and locate faulty device.
- Supports the status statistics of each device, which is convenient for users to quickly understand the running status of each device.
- Supports quick search of target device based on device name.
- You can switch among three modes: large card, small card, and list, and zooming is supported for large card-based and small card-based modes.
- You can click  to edit device indicators, icon color, and style displayed in the device list.
- The large and small card mode supports the token operation of each device, as well as the number of attached device objects to count and filter.
- You can click a device card to enter the corresponding single device monitoring page.

3.1.4.1 Single device monitoring

The single device monitoring mainly covers subsystems, converters, battery arrays, battery clusters, air conditioners, protection measuring-control and liquid cooling devices, which are automatically displayed according to the actual type of device connected to the site. The details of a single device are generally distributed as follows:

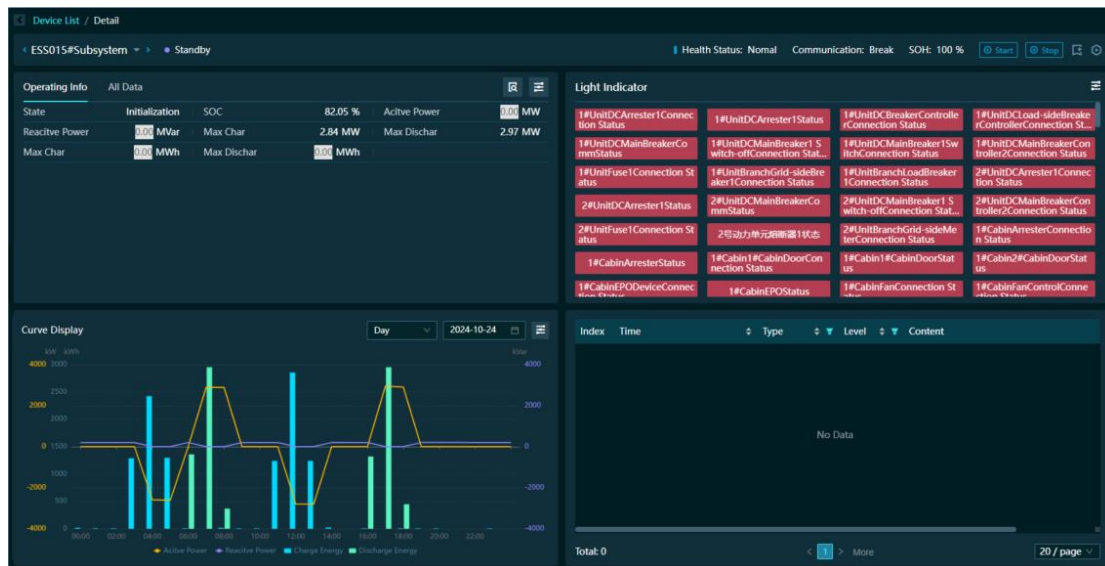


Figure 9: Single device details

The main functions are described as follows:

- You can click "List" to return to the device list page.
- You can click to quickly switch to the previous or next device to improve inspection efficiency, or you can directly click the device name to appear in the asset tree to quickly switch between any device.
- Displays important information of devices, such as real-time running status, health status, and communication status.
- Supports remote start and stop operations of some devices.
- Supports the token operation, the device will automatically block the alarms with the token.
- Displays the operation information. You can click "☰" to set the displayed measurement points and the display style; Click "🔍" to pop up the window to enlarge the display box, and you can view more real-time data of measurement points. "All Data" support the search function.
- Display the light signs, you can click the icon to "update flashing", "return to hide" and "top level" configuration, etc.
- Supports the display of important measurement point curves with different time granularity, such as active power, reactive power, subsystem charging and discharging. Also, the color, style, range and other attributes of curves are customizable.

- Displays the real-time alarm information, and supports the filtration by alarm type and alarm level.

Some devices will display personalized functions according to the needs of actual monitoring. The following is the detailed introduction to various devices.

3.1.4.2 Subsystem

The subsystem list displays all the subsystems at the site and their key operating indicators. Also, you can click a card or a list line to enter the corresponding subsystem detail page.



Figure 10: Subsystem device list

The main functions are described as follows:

- Displays the operating status, the health status and the quantity statistics of various states for all subsystems. By default, the filter bar displays the operating status, including fault shutdown, user shutdown, maintenance and repair, discharge operation, charging operation, equipment standby and connection interruption status. The health status is displayed in the upper left corner of each card below, including normal, alarm, and fault.
- You can click any card or list line to enter its corresponding detail page. The detail page shows the operating status and real-time running data of the device.

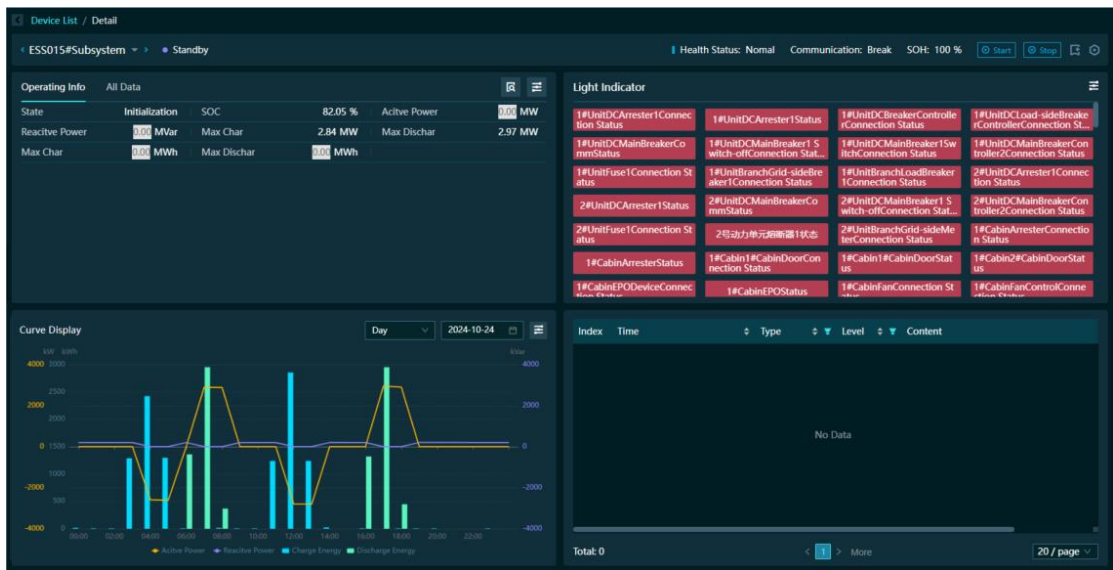


Figure 11: Subsystem details

3.1.4.3 Converter

The converter list displays all the converters at the site and their key operating indicators. Also, you can click a card or a list line to enter the corresponding converter detail page.

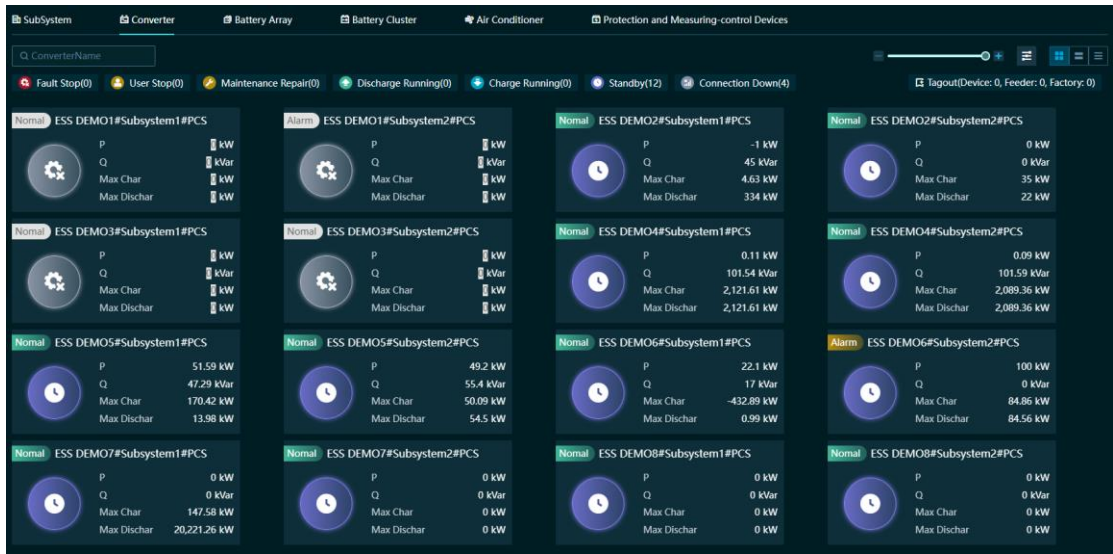


Figure 12: Converter device list

The main functions are described as follows:

- Displays the operating status, the health status and the quantity statistics of various states for all converters. By default, the filter bar displays the operating status, including fault shutdown, user shutdown, maintenance and repair, discharge operation, charging operation, equipment standby and connection interruption status. The health status is displayed in the upper left corner of each card below, including normal, alarm, and fault.

- You can click any card or list line to enter its corresponding detail page. The detail page shows the operating status and real-time running data of the device.

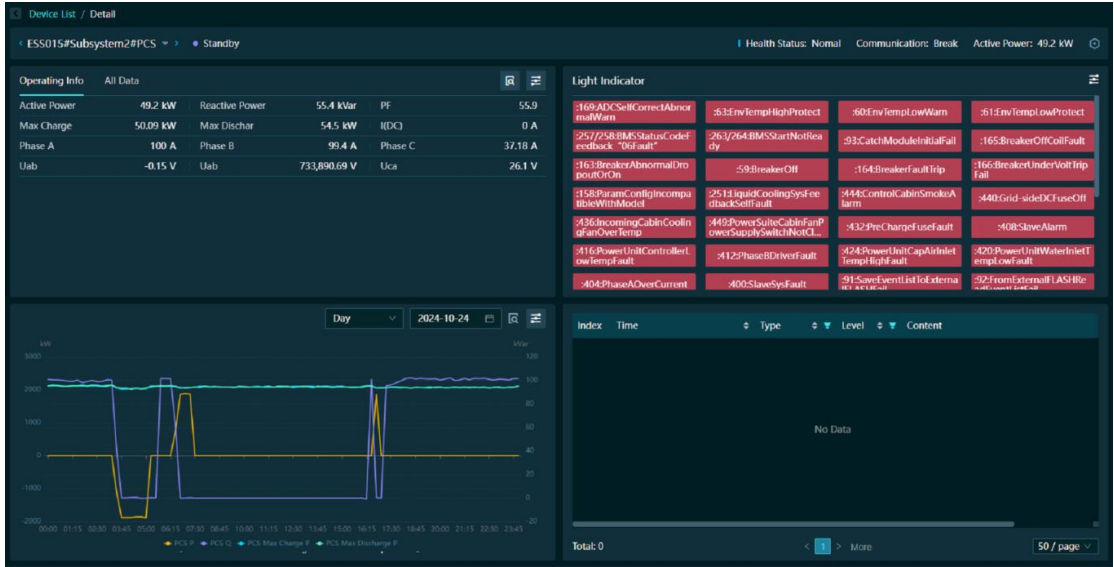


Figure 13: Converter details

3.1.4.4 Battery Array

The Battery Array list displays all the battery arrays at the site and their key operating indicators. Also, you can click a card or a list line to enter the corresponding battery array detail page.

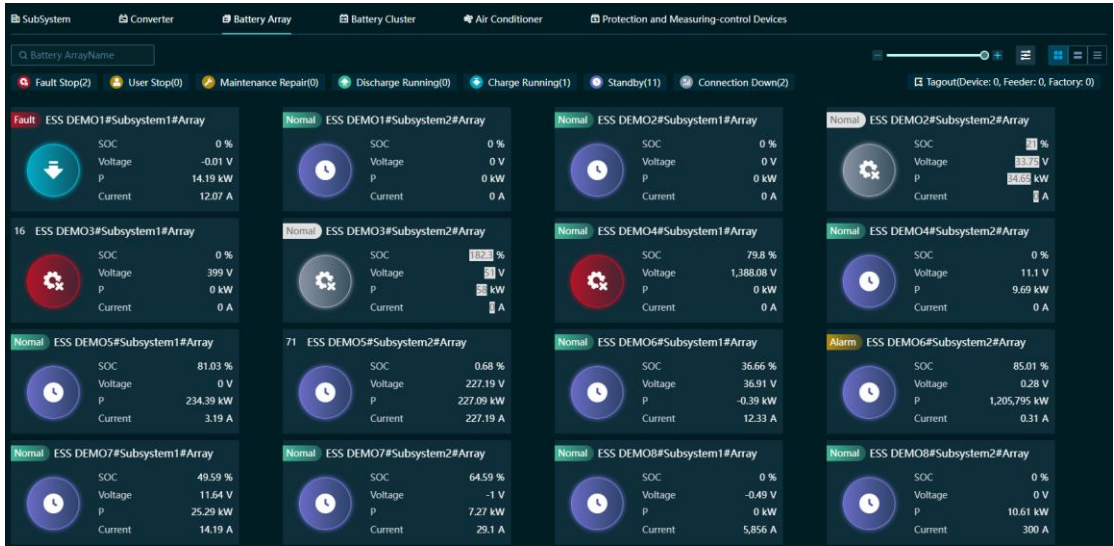


Figure 14: Battery array device list

The main functions are described as follows:

- Displays the operating status, the health status and the quantity statistics of various states for all battery arrays. By default, the filter bar displays the operating status, including fault shutdown, user shutdown, maintenance and repair, discharge operation,

charging operation, equipment standby and connection interruption status. The health status is displayed in the upper left corner of each card below, including normal, alarm, and fault.

- You can click any card or list line to enter its corresponding detail page. The detail page shows the operating status and real-time running data of the device.

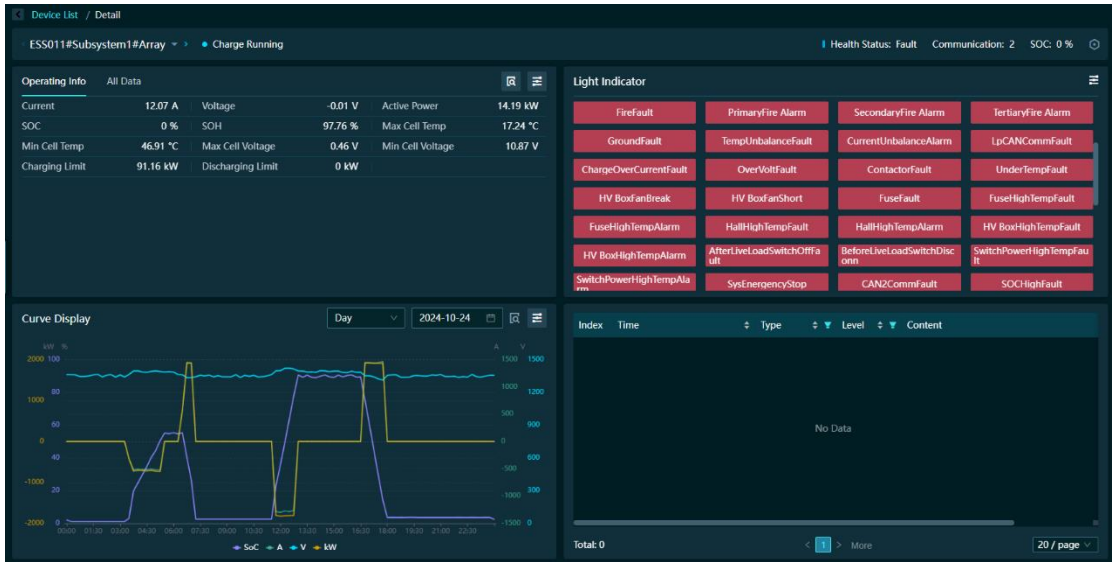


Figure 15: Battery array details

3.1.4.5 Battery Cluster

The Battery Cluster list displays all the battery clusters at the site and their key operating indicators. Also, you can click a card or a list line to enter the corresponding battery cluster detail page.



Figure 16: Battery cluster device list

The main functions are described as follows:

- Displays the operating status, the health status and the quantity statistics of various states for all battery clusters. By default, the filter bar displays the operating status, including fault shutdown, user shutdown, maintenance and repair, discharge operation, charging operation, equipment standby and connection interruption status. The health status is displayed in the upper left corner of each card below, including normal, alarm, and fault.
- You can click any card or list line to enter its corresponding detail page. The detail page shows the operating status and real-time running data of the device.

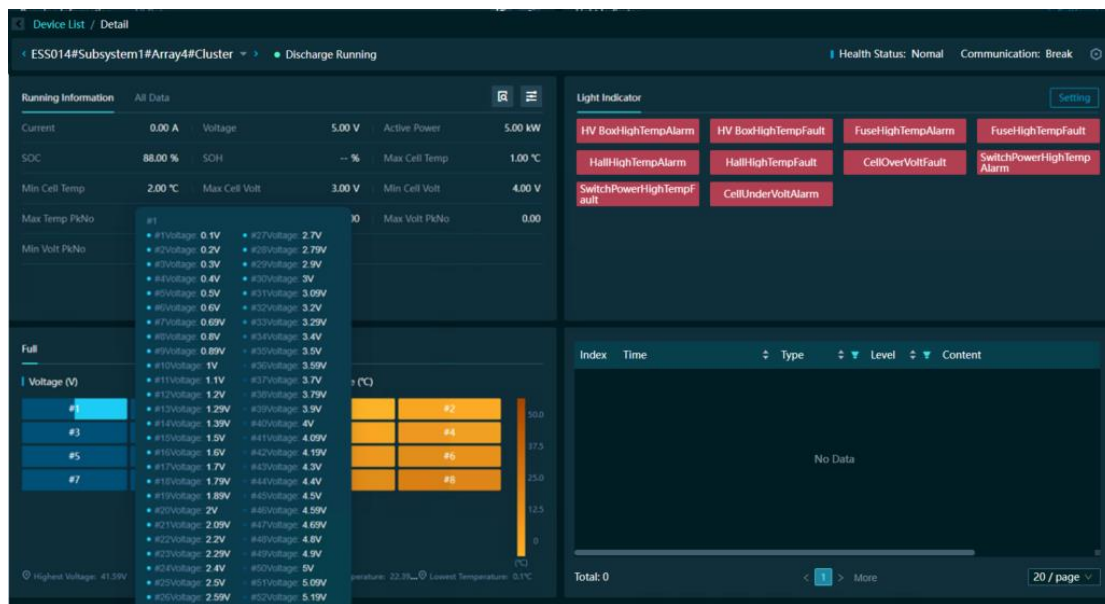


Figure 17: Battery cluster details

The personalized functions of the battery cluster detail page are described as follows:

- Supports the visual display of full cell voltage and temperature under a battery cluster, where the cell voltage and temperature values are distinguished according to different colors, and the position of the highest/lowest cell temperature/voltage can be located.

3.1.4.6 Air Conditioner

The Air Conditioner list displays all the air conditioners at the site and their key operating indicators. Also, you can click a card or a list line to enter the corresponding air conditioner detail page.

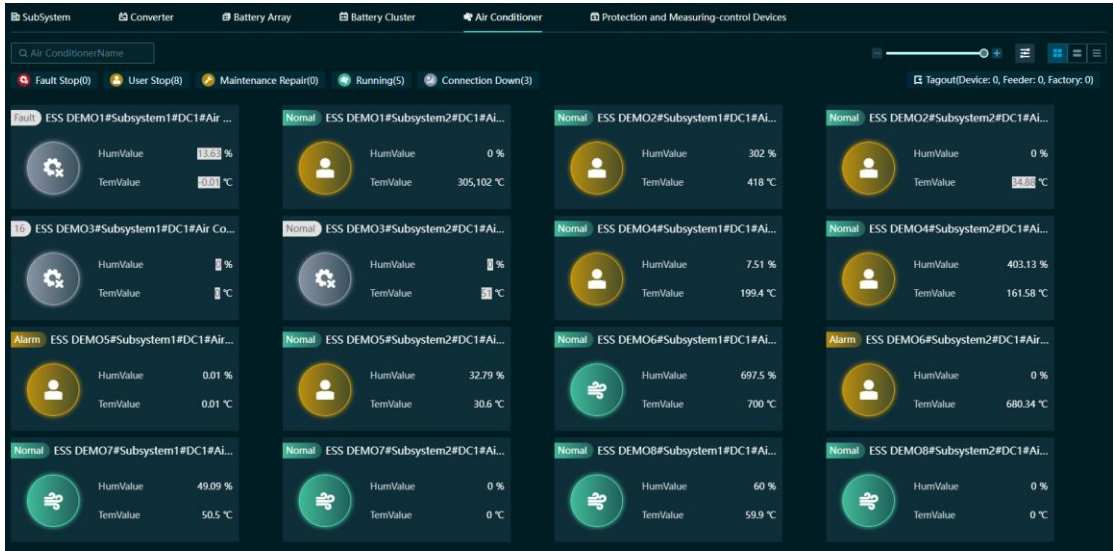


Figure 18: Air conditioner device list

The main functions are described as follows:

- Displays the operating status, the health status and the quantity statistics of various states for all air conditioners. By default, the filter bar displays the operating status, including fault shutdown, user shutdown, maintenance and repair, running, and connection interruption status. The health status is displayed in the upper left corner of each card below, including normal, alarm, and fault.
- You can click any card or list line to enter its corresponding detail page. The detail page shows the operating status and real-time running data of the device.

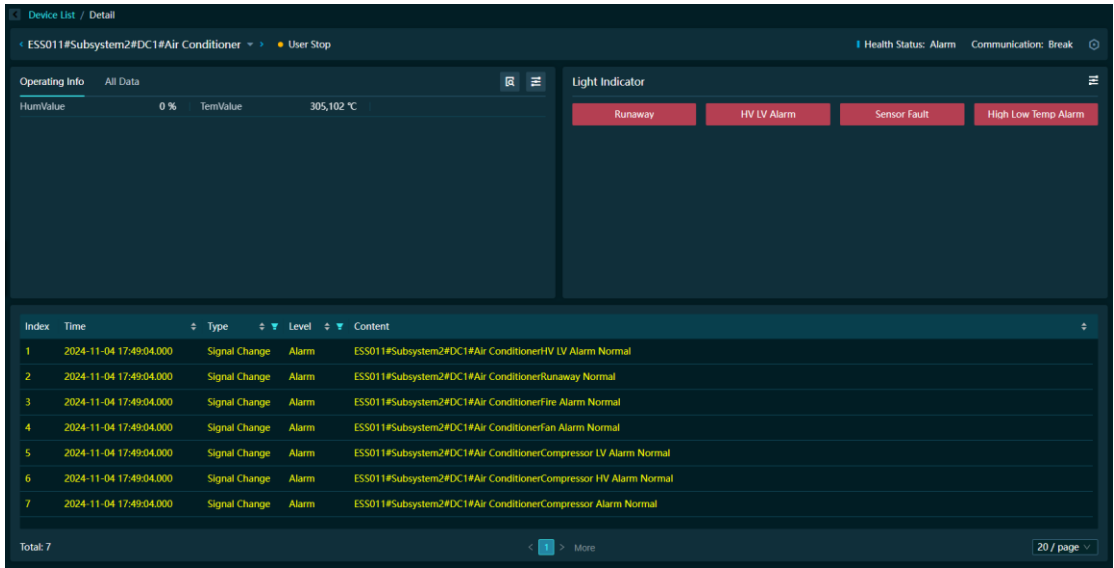


Figure 19: Air conditioner details

3.1.4.7 Protection and Measuring-control Devices

The Protection and Measuring-control Devices device list displays all the measurement and control devices at the site and their key operating indicators. Also, you can click a card or a list line to enter the corresponding protection and measuring-control devices detail page.

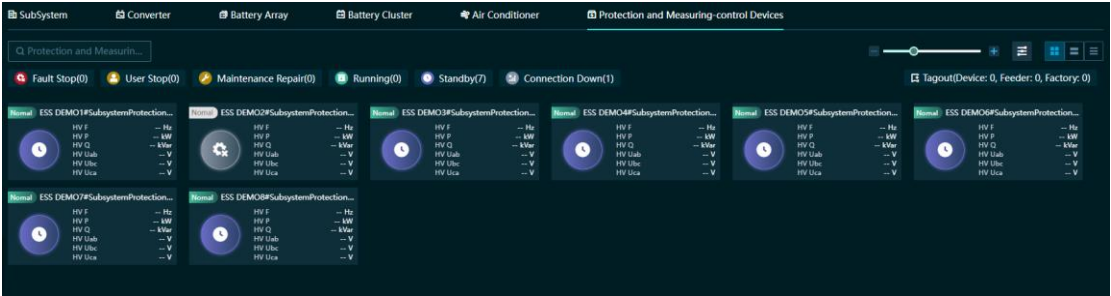


Figure 20: Protection and measuring-control device list

The main functions are described as follows:

- Displays the operating status, the health status and the quantity statistics of various states for all protection and measuring-control devices. By default, the filter bar displays the operating status, including fault shutdown, user shutdown, maintenance and repair, running, standby and connection interruption status. The health status is displayed in the upper left corner of each card below, including normal, alarm, and fault.
- You can click any card or list line to enter its corresponding detail page. The detail page shows the operating status and real-time running data of the device.

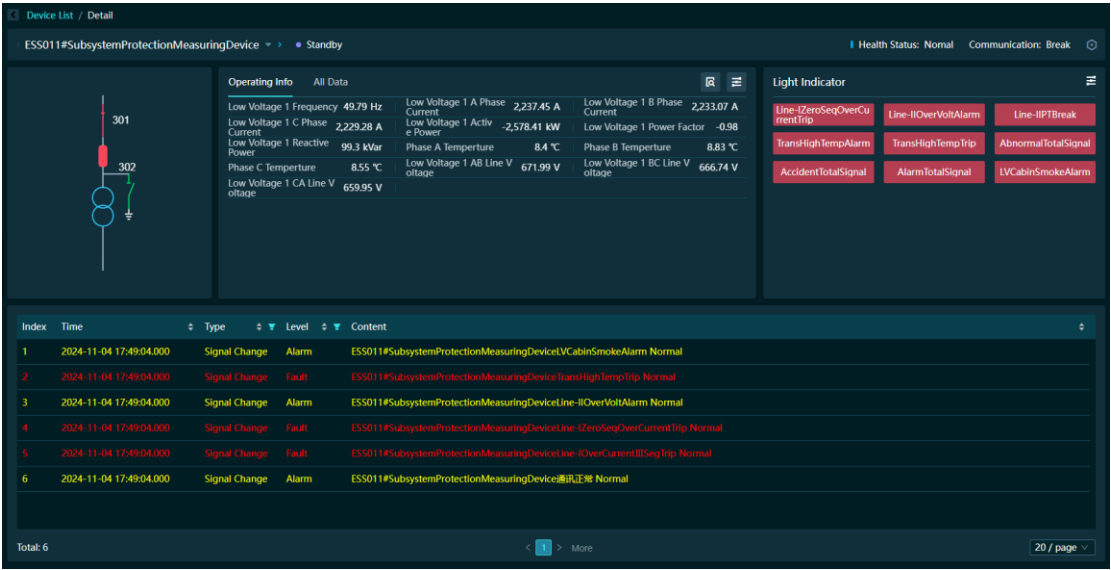


Figure 21: Protection and measuring-control device details

The personalized functions of the protection and measuring-control device detail page are described as follows:

- Supports the display of the primary wiring diagrams of the protection and measuring-control devices, support editing and displaying the device numbers such as switches and circuit breakers.

3.1.4.8 Liquid cooling

The liquid cooling device list displays all the DC liquid cooling devices at the site and their key operating indicators. Also, you can click a card or a list line to enter the corresponding DC liquid cooling devices detail page.



Figure 22: Liquid cooling device list

The main functions are described as follows:

- Displays the operating status, the health status and the quantity statistics of various states for all protection and measuring-control devices. By default, the filter bar displays the operating status, including fault shutdown, user shutdown, maintenance and repair, cooling running, heater running, circulation running and connection interruption status. The health status is displayed in the upper left corner of each card below, including normal, alarm, and fault.
- You can click any card or list line to enter its corresponding detail page. The detail page shows the operating status and real-time running data of the device.

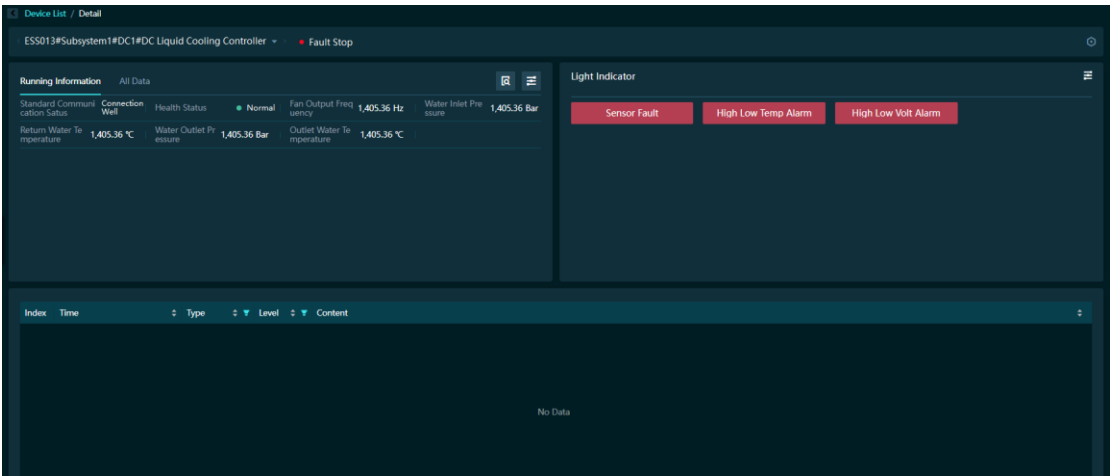


Figure 23: Liquid cooling device details

3.1.4.9 DC Converter

The DC Converter device list displays all the DC Converter devices at the site and their key operating indicators. Also, you can click a card or a list line to enter the corresponding DC Converter devices detail page.



Figure 24: DC Converter device list

The main functions are described as follows:

- Displays the health status, total fault and real-time running data. The health status is displayed in the upper left corner of each card below, including normal, alarm, and fault.
- You can click any card or list line to enter its corresponding detail page. The detail page shows the operating status and real-time running data of the device.

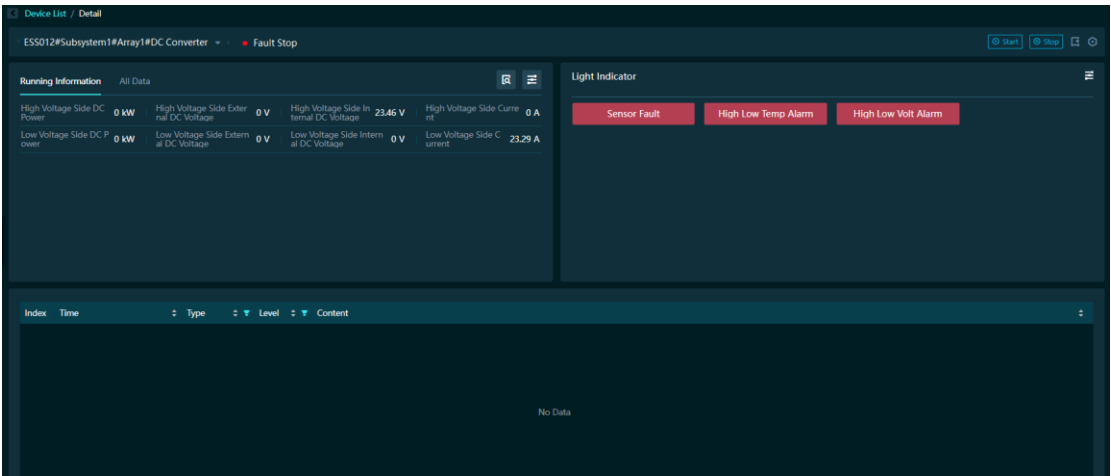


Figure 25: DC Converter device details

3.1.4.10 Other Devices

The list of other devices displays all other non-standard equipment that is not of particular concern under the station, such as dehumidifiers, etc., to realize the monitoring of all connected devices under the station. You can click on the card to jump to the corresponding device details screen.

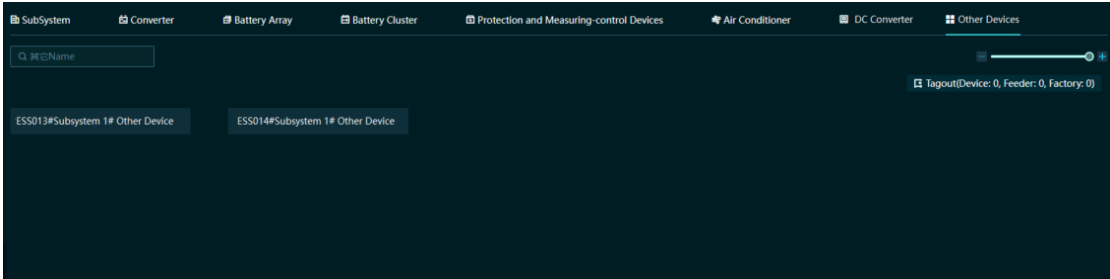


Figure 26: Other devices list

The details page shows the real-time operating data and alarm of the device:

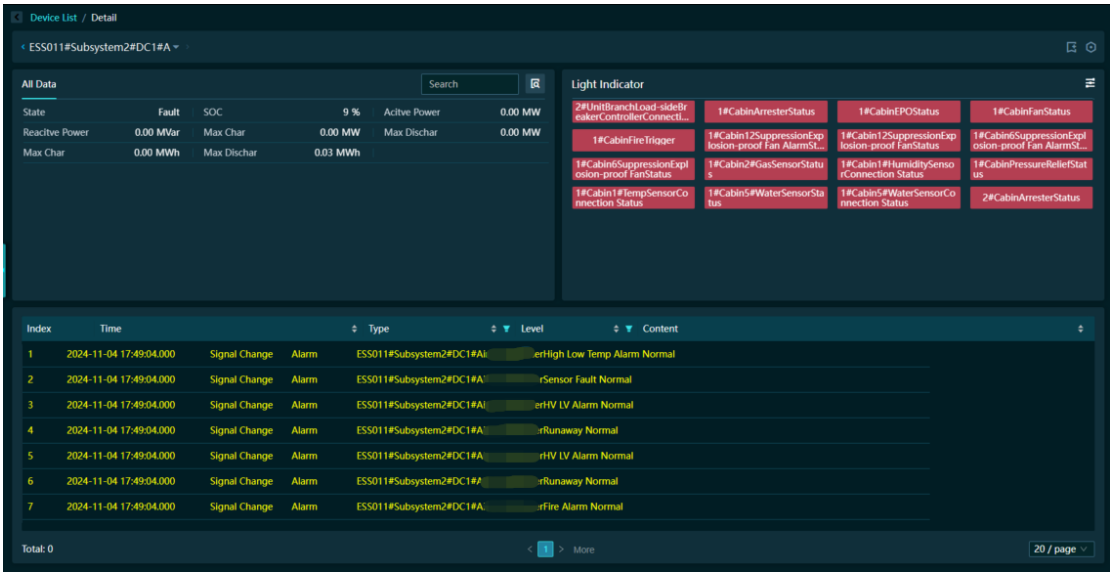


Figure 27: Other devices details

3.1.5 Token List

The token list table page displays all the records of the monitoring object in the station based on the pages of the device list or single device detail, and the tokens can be carried out. Click on the device object to drill down to the details page of the corresponding single device, view the relevant operation information, and confirm the corresponding listing operation.

	Icon	Object	Type	Time	Function	Comment	User	Operation
1	⚠️	ESS01#ESS01	Grounding tag	2024-12-05 01:36:18	lock yk		test	☒ Cancel
2	⚠️	ESS01#ESS01	Test tag	2024-12-05 01:36:19	lock yk		test	☒ Cancel
3	⚠️	ESS011#Subsystem1#Array2#Cluster	Grounding tag	2024-12-05 01:36:24	lock yk		test	☒ Cancel
4	⚠️	ESS011#Subsystem1#Array2#Cluster	Danger tag	2024-12-05 01:36:25			test	☒ Cancel
5	⚠️	ESS011#Subsystem2#Array4#Cluster	Grounding tag	2024-12-05 01:36:30	lock yk		test	☒ Cancel
6	⚠️	ESS011#Subsystem2#Array9#Cluster	Danger tag	2024-12-05 01:36:39			test	☒ Cancel

Figure 28: Token List

3.1.6 Favorite

When the on-duty personnel monitor the system, they may pay attention to multiple monitoring pages. At present, most users rely on multiple monitoring screens to solve the problem. In order

to improve the convenience of switching back and forth in the case of more monitoring screens, the system provides the favorite and rotate functions, and the pages in the favorite can automatically redirect in turn to improve the efficiency of user monitoring. The main functions are:

- Click "★ Favorite" in the upper right corner, customize the page tag name and save, that can be added the current page to the favorite list.

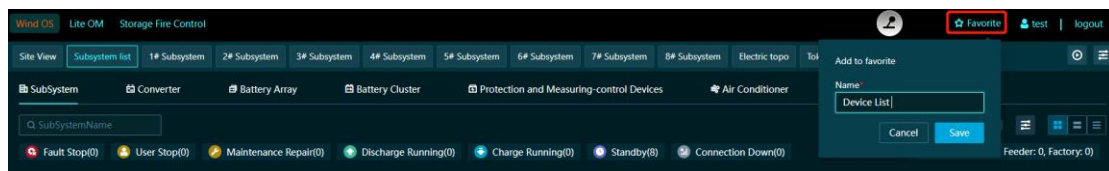


Figure 29: Add favorite

- Display all the page tags that have been added to the favorites bar at the top of the page, and you can click the page tag to quickly jump to the corresponding page. On the right, you can start/pause the page rotation and set the page labels.

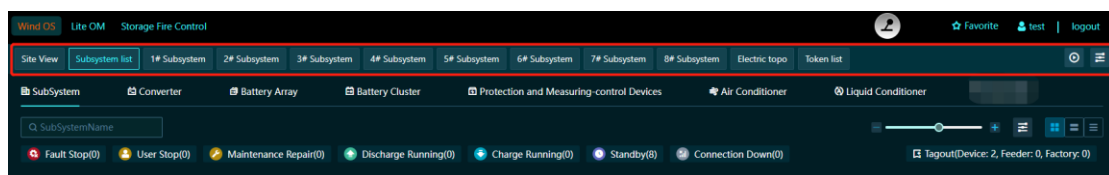


Figure 30: Favorite list

- Click the settings button on the right to modify the name, delete, whether to rotate, and set the rotation duration of the tags that have been added to the favorite page.

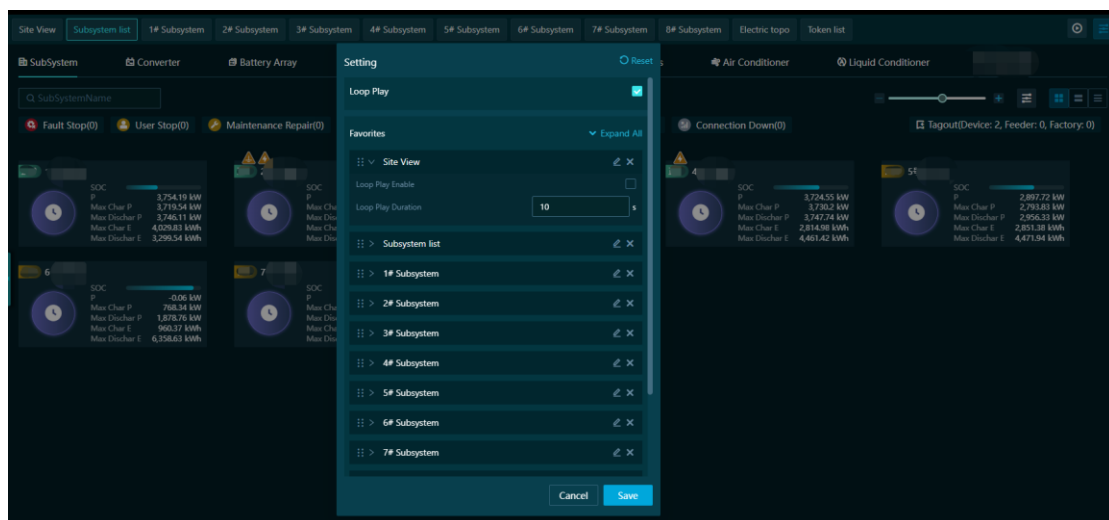


Figure 31: Favorite setting

3.2 Alarm

3.2.1 Real-time alarm

The site control system operating and alarm information is displayed in real time on the real-time alarm page. The display style of alarm information in the page can be customized by the user on the alarm template configuration page. After the settings are completed, the real-time alarm interface will display the alarm information according to the set template. On the real-time alarm page, users can filter, sort and confirm the alarm information, and can filter the alarm information by confirmation status, alarm type, alarm level, field, application and activity. The alarm types supported by the system include: remote signal displacement, remote signal SOE, telemetry limit violation, working condition information, manual operation, remote control and remote adjustment, protection events, etc. For remote signal displacement and fault (SC) types of alarms, users can configure and modify records on the interface. For the real-time alarms, the alarms at levels below the interval are given by application.

If the frequency of a specific type of alarm from a specific device exceeds a predefined threshold within a certain time window, the historical alarms will be removed from the database to reduce storage load, while still retaining full traceability via the archived CSV files.

Once the alarm frequency exceeds the configured threshold, the system will also generate a summarized archival alarm, such as:

“[Alarm Type] occurred [X] times from [Start Time] to [End Time] at [Object].”

This helps operators quickly understand the context and volume of repetitive alarm events without reviewing each instance individually.

Filter										Stop	Confirm All	Collapse
Confirm :		Confirmed	Unconfirmed			Level :		Information	Alarm	Fault		
Type :	Signal Change	Remote Control	Relay Event	Status Code	Connection Status							
		Time	Site	Device	Type	Level	Confirmed	Content				
1		2022-11-14 13:14:31.168(UTC+08:00)		System	System Information	Information	Yes	demo [User Sign In]--10.81.141.48--main1				
2		2022-11-14 13:14:20.839(UTC+08:00)		System	System Information	Information	Yes	demo [User login failed]--10.81.141.48--main1				
3		2022-11-14 12:14:38.019(UTC+08:00)		System	System Information	Information	Yes	demo [User Session Time Out]--10.100.193.234--main1				
4		2022-11-14 11:54:38.021(UTC+08:00)		System	System Information	Information	Yes	demo [User Session Time Out]--10.100.193.72--main1				
5		2022-11-14 11:32:27.037(UTC+08:00)		System	System Information	Information	Yes	demo [User Sign In]--10.100.193.72--main1				
6		2022-11-14 11:20:50.663(UTC+08:00)		System	System Information	Information	Yes	demo [User Sign In]--10.100.193.234--main1				

Figure 32: Real-time alarms

You can configure the pop-up alarm configuration on the page according to the alarm template, and pop up a single pop-up window or a double pop-up window (the fault pop-up window and the alarm pop-up window appear separately and independently).

Based on historical alarm data, the system supports more advanced alarm analysis functions. You can click the button with "Analyse" in the "Operate" column to enter the corresponding analysis function interface. At present, the energy storage station supports the frequency analysis function.

Filter								Stop	Confirm All	Collapse
Confirm :		Confirmed	Unconfirmed	Level :			Information	Alarm	Fault	
Type :	Signal Change	Remote Control	Relay Event	Status Code	Connection Status					
		Time	Site	Device	Type	Level	Confirmed	Content	Operate	
1	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
2	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
3	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
4	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
5	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
6	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
7	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
8	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
9	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
10	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
11	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
12	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
13	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
14	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
15	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
16	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
17	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
18	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	
19	🔍	12/04/2024 PM 07:36:14	F02	F02-A10	Signal Change	Information	No	F02-A10Downline state Alarm	Analyse	

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Figure 33: Alarm analysis

3.2.1.1 Frequency analysis

Frequency analysis mainly includes longitudinal analysis and horizontal analysis. Based on the analysis results, you can perform operations such as not alarming, modifying the alarm level, and disabling voice. The main functions are:

- **Longitudinal analysis:** You can analyze the frequency of alarms every day in the past week. Analyze the number of alarms of the same type on the same device to reflect the running status of the device.

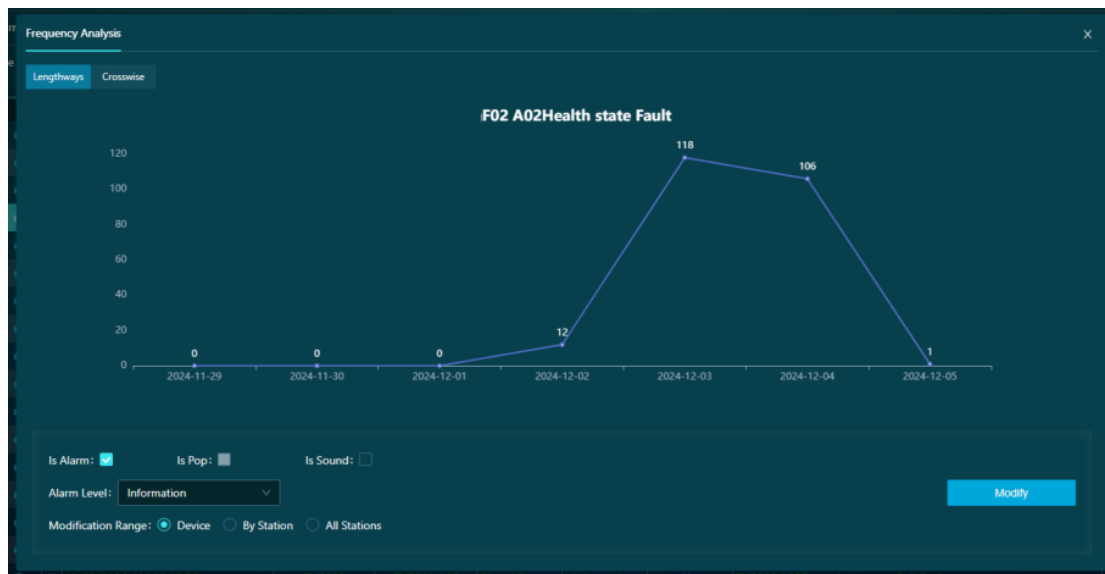


Figure 34: Longitudinal analysis

- Horizontal analysis: Supports selecting alarms for the same type of devices at the same time. By analyzing the number of alarms of different devices of the same type, the differences in operation between devices of the same type are reflected.

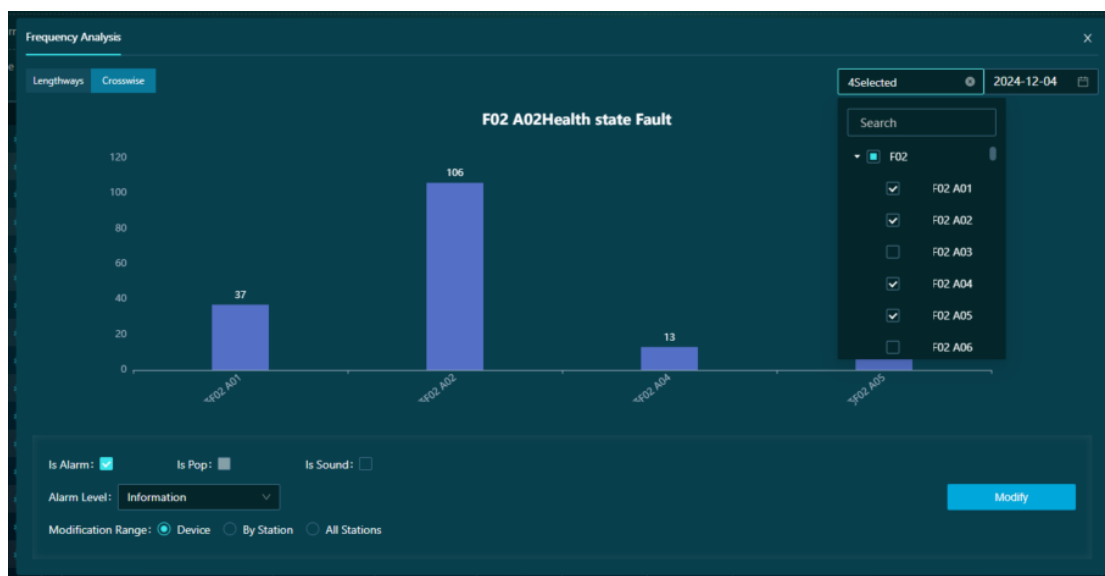


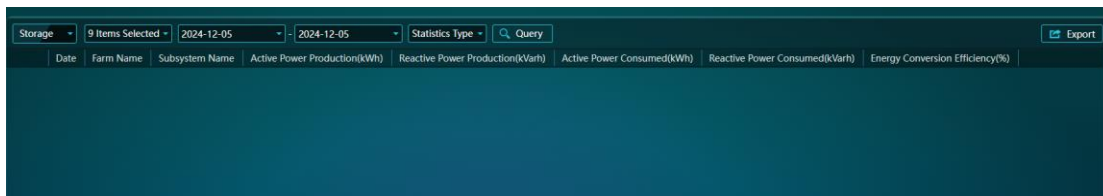
Figure 35: Horizontal analysis

3.2.2 Historical alarm

The historical alarm function is mainly used to search various alarm information that has occurred. Users can filter and query alarms by alarm type, device, alarm time, and alarm level. The user can save the query conditions as a query template to facilitate the next query for the same type and level of alarm information. The query results can be exported into CSV, XLS, PDF and other formats and saved to the specified path.

3.3 Report

The site control system module provides the energy reports in the field of energy storage. The statistics of daily, monthly and annual energy are made from two dimensions, i.e. energy storage site and energy storage subsystems, including active and reactive discharge/discharge power and energy conversion efficiency. Users can also export and save the data of energy storage power report as CSV, XLS or PDF files according to actual needs.



3.3.2 Operation report

39

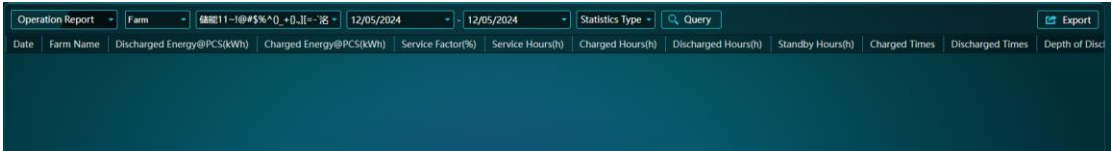


Figure 38: Operation report

3.3.3 TBA report

The station control system module provides a utilization report in the field of energy storage. Supports daily, weekly, monthly, and annual statistics for stations and subsystems within a specified time range. At the same time, it provides the function of editing the shutdown record for external reasons for the device type to solve the problem that the availability of the equipment itself and the external cause is lower than the actual utilization rate.

3.3.3.1 TBA

You can query and export the availability of a site/subsystem within a specified time range, such as daily, weekly, monthly, or yearly.

Availability Report		Device Shutdown Record	
Farm	2024-10-10 -- 2024-10-27	Day	Inquire
Date		Farm Name	Availability
1	2024-10-10	储能11	19.722%
2	2024-10-11	储能11	0.000%
3	2024-10-12	储能11	0.000%
4	2024-10-13	储能11	0.000%
5	2024-10-14	储能11	28.755%
6	2024-10-15	储能11	37.037%
7	2024-10-16	储能11	37.037%
8	2024-10-17	储能11	37.037%
9	2024-10-18	储能11	37.037%
10	2024-10-19	储能11	37.037%
11	2024-10-20	储能11	37.037%
12	2024-10-21	储能11	37.037%
13	2024-10-22	储能11	37.037%
14	2024-10-23	储能11	25.833%
15	2024-10-24	储能11	37.037%
16	2024-10-25	储能11	25.833%
17	2024-10-26	储能11	25.833%

Figure 39: TBA report

3.3.3.2 Downtime record

You can query subsystem-oriented downtime records, edit downtime and reason, and export downtime record data within a specified time range. At the same time, it supports the export of 1min-level reporting reports, power supply network data reporting, and other needs.

	Start Time	End Time	Subsystem Name	Subsystem	Converter	Battery Array	Protection...	ECO Reason	ECO Factor
☐	1	10/14/2024 06:02:00		19	0.000	0.000	0.000	0.000	0.000
☐	2	10/14/2024 06:01:00	10/14/2024 06:02:00	19	0.000	1.000	0.000	0.000	0.000
☐	3	10/14/2024 06:00:00	10/14/2024 06:01:00	19	0.000	2.000	0.000	0.000	0.000
☐	4	10/14/2024 05:55:00	10/14/2024 06:00:00	19	0.000	1.000	0.000	0.000	0.000
☐	5	10/14/2024 05:48:00	10/14/2024 05:55:00	19	0.000	2.000	0.000	0.000	0.000
☐	6	10/14/2024 05:38:00	10/14/2024 05:48:00	19	0.000	1.000	0.000	0.000	0.000
☐	7	10/10/2024 04:44:00	10/14/2024 05:38:00	19	0.000	0.000	0.000	0.000	0.000
☐	8	10/01/2024 09:38:00	10/10/2024 04:44:00	19	0.000	2.000	0.000	0.000	0.000

Figure 40: Downtime record

Click the button next to the end time  to quickly split the shutdown record. In addition, you can check multiple shutdown records, click the "Batch Edit" button to edit the availability and reason of the shutdown record subsystem, and explain the situation in the remarks.

3.4 Analysis

3.4.1 Trend analysis

The trend analysis tool can reflect the change trend of the telemetry value, the remote signal value and the electrical measurement in the form of tables and curves. Users can compare the historical values of measurement points in the form of curves or tables, and can query the maximum and minimum historical values of telemetry points. The system supports simultaneous display and selective cross-comparison of the curves of different colors for multiple measurement points as well as the zooming-in/zooming-out of curves. The searched records can be exported to the specified path in CSV, XLS, PDF and other formats. The main functions are as follows:

- Customization of consistency analysis templates based on device type and model is supported. For example, it supports battery pack differential pressure analysis, temperature differential analysis, and current analysis by default; Cell pressure difference analysis, temperature difference analysis, etc., you can directly select the corresponding template for query and analysis within the time range needed.
- The maximum, minimum, and average statistics of the queried data can be performed.
- Data can be queried by day, week, month, year, or custom time range.
- Supports year-on-year and month-on-month analysis in the corresponding time dimension.

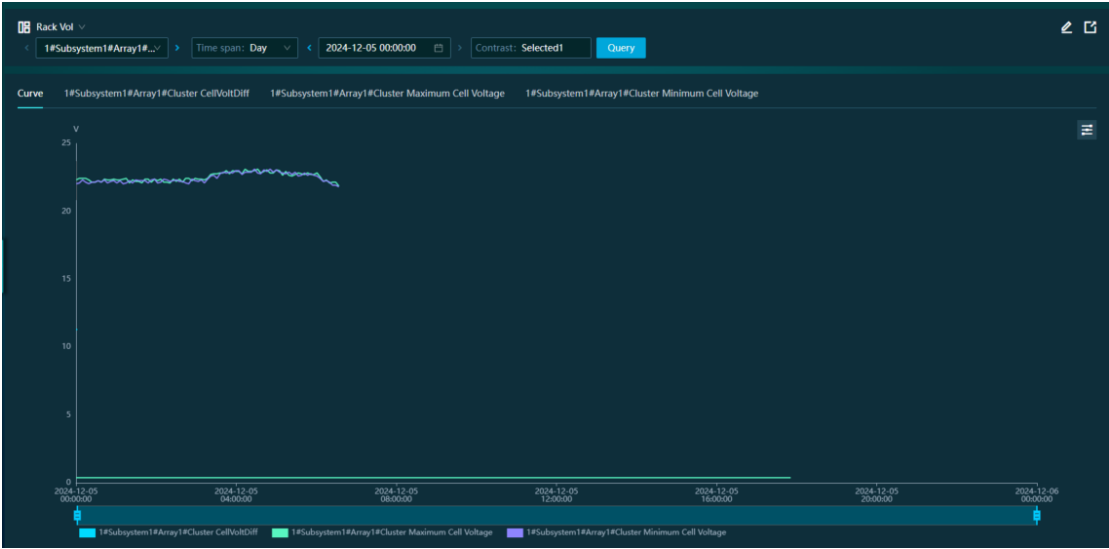


Figure 41: Trend graph curve

New Query

Condition

1. Select query scope

Domain: storage domain

Type: Rack

Model: Please select

2. Select assets

Selected1

ESS01

1#Subsystem1#Array1#CL...

1#Subsystem1#Array2#CL...

1#Subsystem1#Array3#CL...

1#Subsystem1#Array4#CL...

1#Subsystem1#Array5#CL...

1#Subsystem1#Array6#CL...

1#Subsystem1#Array7#CL...

1#Subsystem1#Array8#CL...

1#Subsystem1#Array9#CL...

1#Subsystem2#Array1#CL...

1#Subsystem2#Array2#CL...

3. Select points

AllType: Please select

BMUEqualizationSt...

BMUEqualizationSt...

BMUEqualizationSt...

BMUEqualizationSt...

BMUEqualizationSt...

BMUEqualizationSt...

BMUEqualizationSt...

BMUEqualizationSt...

Total Current

AlarmClearValue-C...

AlarmClearValue-H...

AlarmTriggerValue...

AlarmTriggerValue...

AlarmTriggerValue...

AlarmTriggerValue...

+ Add condition

Time: Time span: Day Please select Select date Contrast: Please select

Data statistics Maximum Maximum Time Minimum Minimum Time Average Sampling Granularity

Save as a template Yes No

Cancel Query

Configuration

Reset

1#Subsystem2#PCS Power Factor

Show Title

Convert + New Convert

Y-axis public Special

Position Left Right

Subtract Please select

Style /

1#Subsystem2#PCS AB Line Voltage

Show Title

Convert + New Convert

Y-axis public Special

Position Left Right

Subtract Please select

Common left axis minimum maximum

Common right axis minimum maximum

Cancel Save

Figure 42: Template and display style editing

Curve	1#Subsystem1#Array1#Cluster Cell VoltDiff	1#Subsystem1#Array1#Cluster Maximum Cell Voltage	1#Subsystem1#Array1#Cluster Minimum Cell Voltage
	Time	Value	State
1	2024-12-05 00:00:00	0.38	Normal
2	2024-12-05 00:09:00	0.38	Normal
3	2024-12-05 00:18:00	0.38	Normal
4	2024-12-05 00:27:00	0.38	Normal
5	2024-12-05 00:36:00	0.38	Normal
6	2024-12-05 00:45:00	0.38	Normal
7	2024-12-05 00:54:00	0.38	Normal
8	2024-12-05 01:03:00	0.38	Normal
9	2024-12-05 01:12:00	0.38	Normal
10	2024-12-05 01:21:00	0.38	Normal

Figure 43: Trend chart table

3.4.2 Scatter analysis

The scatter analysis can reflect the change trend of the telemetry value in the form of tables and curves. Users can compare the historical values of measurement points in the form of curves or tables. The system supports display and selective cross-comparison for multiple measurement points as well as the zooming-in/zooming-out of curves. The searched records can be exported to the specified path in CSV, XLS, PDF and other formats.

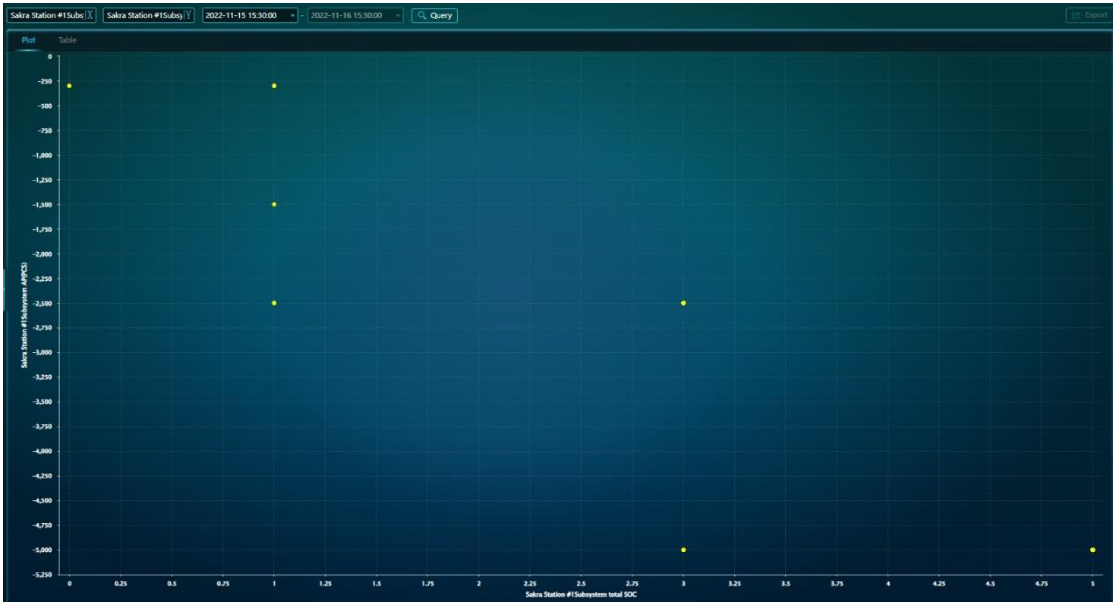


Figure 44: Scatter analysis curve

	Time	Sakra Station #1 Subsystem total SOC (X)	Sakra Station #1 Subsystem APPCS (Y)
13	2022-11-15 15:42:00	3	-2,500
14	2022-11-15 15:43:00	3	-2,500
15	2022-11-15 15:44:00	3	-2,500
16	2022-11-15 15:45:00	3	-2,500
17	2022-11-15 15:46:00	3	-2,500
18	2022-11-15 15:47:00	3	-2,500
19	2022-11-15 15:48:00	3	-2,500
20	2022-11-15 15:49:00	3	-2,500
21	2022-11-15 15:50:00	3	-2,500
22	2022-11-15 15:51:00	3	-2,500
23	2022-11-15 15:52:00	3	-2,500
24	2022-11-15 15:53:00	3	-5,000
25	2022-11-15 15:54:00	3	-5,000
26	2022-11-15 15:55:00	3	-5,000
27	2022-11-15 15:56:00	3	-5,000
28	2022-11-15 15:57:00	5	-5,000
29	2022-11-15 15:58:00	5	-5,000
30	2022-11-15 15:59:00	5	-5,000
31	2022-11-15 16:00:00	5	-5,000
32	2022-11-15 16:01:00	5	-5,000
33	2022-11-15 16:02:00	5	-5,000
34	2022-11-15 16:03:00	5	-5,000
35	2022-11-15 16:04:00	5	-5,000
36	2022-11-15 16:05:00	5	-5,000
37	2022-11-15 16:06:00	5	-5,000
38	2022-11-15 16:07:00	5	-5,000
39	2022-11-15 16:08:00	5	-5,000
40	2022-11-15 16:09:00	5	-5,000
41	2022-11-15 16:10:00	5	-5,000
42	2022-11-15 16:11:00	5	-5,000

Figure 45: Scatter analysis table

3.4.3 Data export

The data export function supports on-demand export of measurement point data and historical alarms, and supports multi-dimensional customized export such as device type, sampling granularity, asset selection, measurement point selection, time range selection, etc., and downloads or deletes. At the same time, this function supports saving export conditions as templates.

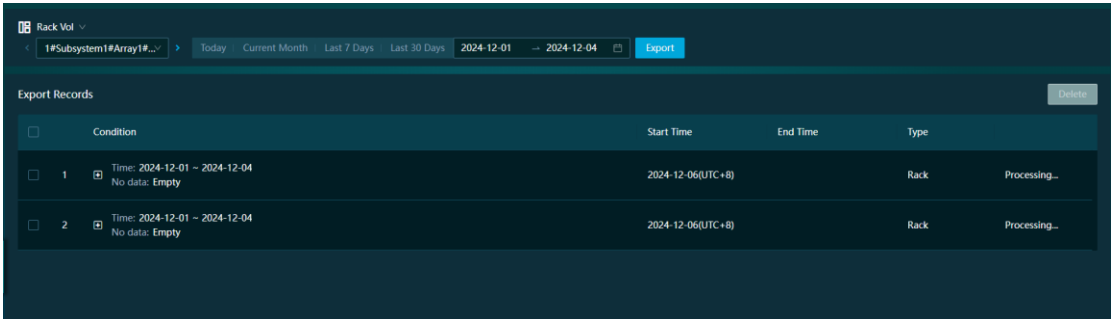


Figure 46: Data export

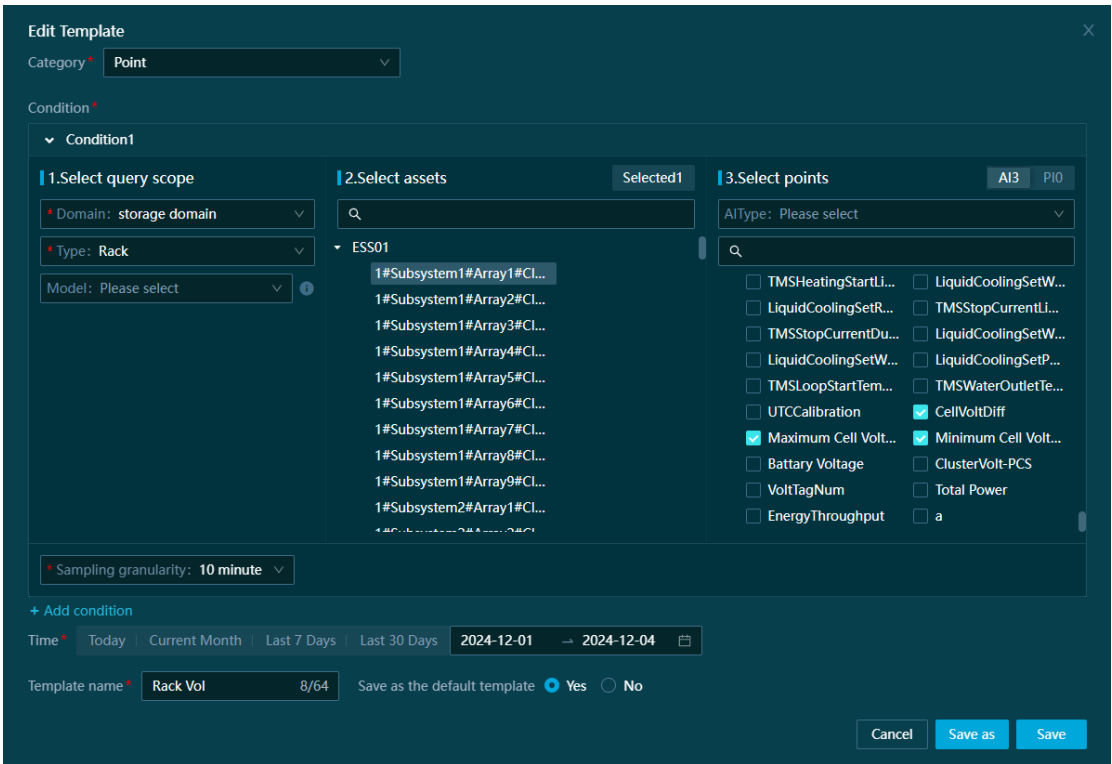


Figure 47: Export template editing

3.5 Setting

The system settings mainly covers three modules: operation parameters, alarm template configuration, and password modification.

- Operating parameters: set the parameters of energy storage operation.

- Alert template configuration: used to customize the real-time alarm template and set the pop-up alarm mode.
- Password modification: used to reset the password of the current login account;

3.5.1 Operating parameters

The alarm template configuration is mainly to provide users with the role of super user with the function of real-time alarm template customization and the function of pop-up alarm mode setting.

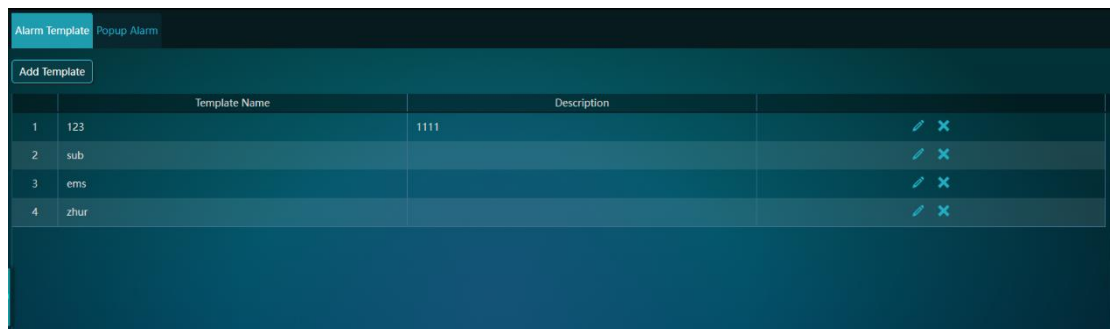


Figure 48: Alert template customization

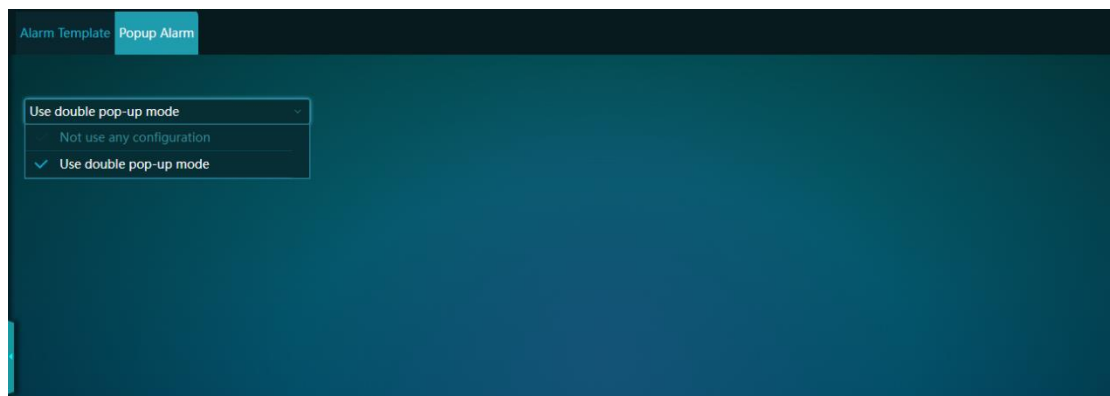


Figure 49: Pop-up alarm mode settings

3.5.2 Password Modify

This part is mainly used to reset the password of the current login account.

Modify User

Basic Information

User Type

Normal User

User Information

User Name: demo

ID: 0000000000

User Alias: demo

User Email: 111@163.com

Account Type: Permanent

User IP: Connect IPs with comma

Priority: Normal

Homepage:

Logo:

Privilege Information

Confirm Cancel

Figure 50: Modify the password

3.6 User management

The user management covers two categories: account management and role management. Only administrators have user management permission.

The account management function mainly refers to the addition, deletion and modification of users, the setting of login homepage, password and LOGO, and the association of account and role permissions. In addition, the super user has the permission to unlock any locked user. If the user name or password is entered incorrectly three times in a row when a user logs into the system, the account will be locked. At this time, the super user can unlock the account through the user unlocking function so that the user can log into the system immediately.

Modify User

Basic Information

User Type

Normal User

User Information

User Name: demo

ID: 0000000000

User Alias: demo

User Email: 111@163.com

Account Type: Permanent

User IP: Connect IPs with comma

Priority: Normal

Homepage:

Logo:

Confirm

Figure 51: User management

User Management

Role

Add Modify Delete Reset Password

	User Name	User Alias	User Email	User Type	User IP	Temporary Account	Priority	Approval State	User State
1	demo	demo	111@163.com	Normal User		No	Normal	Approved	Normal
2	sagl	sagl	sagl@sagl.com	Normal User		No	Normal	Approved	Unlock
3	admin	admin	163@163.com	Normal User		No	Normal	Approved	Unlock
4	normal	normal	normal@1.com	Normal User		No	Normal	Approved	Unlock

Figure 52: User unlocking function

The role management function is mainly to establish corresponding job roles according to business needs. At the same time, these positions can be cross-configured with business fields, which means strong role establishment flexibility.

For example, if the roles of system administrator, duty officer, duty leader are established, the roles of wind farm duty officer, solar duty leader and booster station visitor can be formed by combining with wind, solar and hydropower, just to name a few. The final role will match the screens of business field, that is, the solar duty leader can only access the corresponding screens of solar site, and cannot access the screens of hydro site or booster station; the booster

station visitor can only access the booster station screens and does not have the operation permissions. The functions of adding roles, modifying role permissions and deleting roles are shown in the figure below.

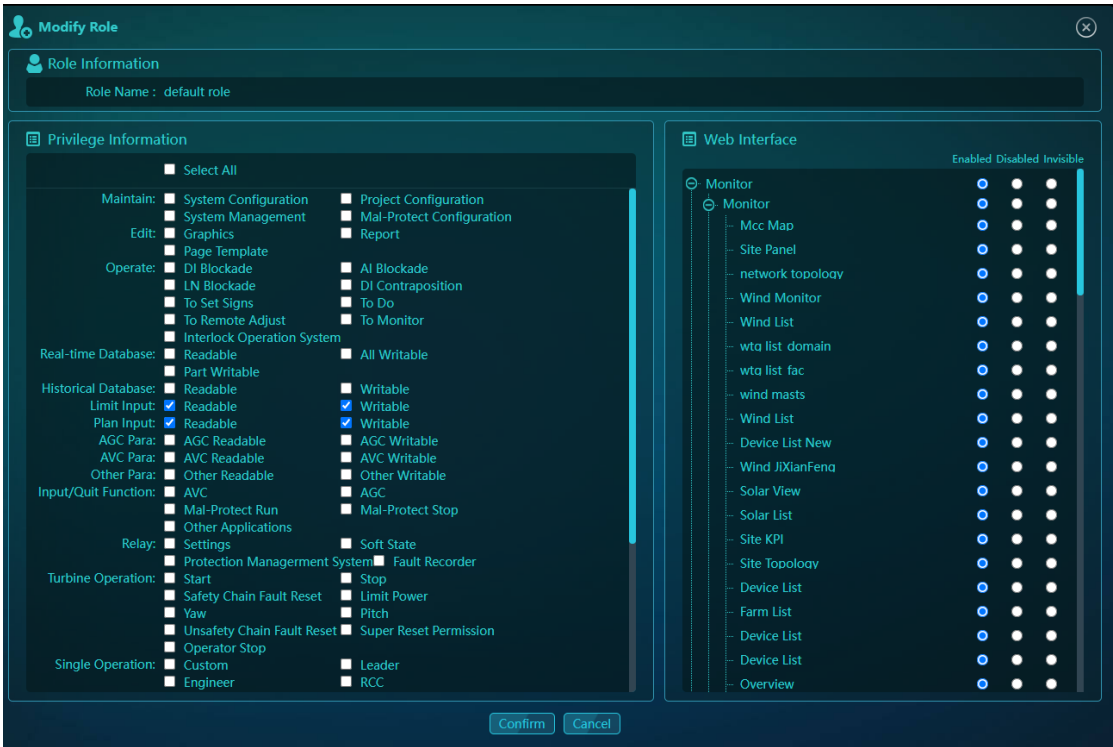


Figure 53: Role management

4 System Performance

4.1 Access capacity

Based on real-time databases, timing databases and general commercial relational databases, the EnOS™ BESS SCADA system is designed with a distributed scheme to organize massive data, create data models, establish data relationships, collect real-time data, and quickly access the historical data. A single node supports a maximum access scale of 200,000 points.

4.2 Data storage

The historical data storage mainly covers 1-min data, event information and report data. The 1-min data and 10-min data are acquired through sampling, and the cycle selected on the data collection device can be used as the shortest storage cycle without the restriction on the number of sampling points. The event information is completely recorded in the historical database to ensure traceability and benchmarking. In addition, the system provides the report data that reflects the operating status of different devices, and stores it in the historical database.

The historical data storage capacity is calculated according to the following table:

Type	Storage period	Capacity increment	Required capacity (for every 100K points)
Historical report	10 years	5 GB/100K points/year	50 GB
1-min data	2 years	5 GB/100K points/year	800 GB
Event information	2 years	5 GB/100K points/year	200 GB

4.3 Performance indicators

- System redundancy indicators

Hot standby switching time	≤ 3s
Cold standby switching time	≤1 min

- System computer resource load rate indicators

Average CPU load rate (within any 5 minutes)	≤40%
Standby space (root zone)	≥20% (or 10G)

- SCADA indicators

Number of accessible points per distributed front-end group	≤ 100000 points
Number of accessible points for a single site control node	≤ 200000 points
Real-time data change update delay	≤ 1 s
Accessible points for remote control	≥ 30000
Master site remote output delay	≤ 1 s
Event recording resolution	≤ 1 ms
Remote signal displacement transmission time	≤ 2 s
One-hour data export	≤ 5 min
Daily report calculation	≤ 5 min
Raw data storage	≤ 2 years

5 Deployment Requirements

5.1 Software environment

The software environment for the system is described as follows:

Software	Version
OS	Linux v6/Ubuntu 22(containerization)
MySQL	v5.7
Chrome	V90.0+

5.2 Hardware Requirement

Processor (CPU): Quad-core processor, 2.1 GHz or higher

Memory (RAM): 32 GB

Storage: 7.2 T

5.3 System security

In order to improve its security, the system has been enhanced for its security protection capabilities from four aspects: permission management, security audit, account security and session management.

5.3.1 Permission management

In order to improve the system security, three independent management roles, including system administrator, audit administrator, and business configurator, are set for the system. Different roles have independent management permissions and management scope.

The role of audit administrator only has the permissions to monitor the operation tracks of other types of users and to manage, monitor and operate & maintain audit records. The audit records mainly cover events, user names, user IPs, audit event types, event levels, and event categories. The audit event types mainly include: manual remote signaling lockout, manual telemetry lockout, manual remote control, manual remote adjustment, manual card placement, user event login, user login/logout, and early warning in audit log. The event levels include hints, alarms and failures. The event categories cover business level and system level, where the system-level events include user event login, user login/logout, and early warning in audit log while the business-level events include manual remote signaling lockout, manual telemetry lockout, manual remote control, manual remote adjustment, and manual card placement.

The role of system administrator only has system management permissions, including user management, role management, permission management and configuration customization. The role of business configurator only has the permissions for basic configuration of various parameters, master data and functions.

The business operators are the end business users of the system, and have no management permissions. Also, the permissions of mutually exclusive services cannot be granted to the same business user.

5.3.2 Security audit

The system performs the security audit on important user behaviors and security events of the operating system and database systems. Such audit covers all the users and the audit records include the event date and time, user, event type, and other audit-related information.

- The system regularly backs up the audit records of the operating systems and database systems, and the audit records cannot be tampered with;
- The application audit records of the system can be exported, classified, sorted, inquired and statistically analyzed

5.3.3 Account security

In order to achieve account security, the system increases constraints on the password complexity and the number of logins.

- Login verification

When a user logs into the system, the system will judge and determine whether the password strength verification is required, whether it is the first login for the user, whether the password has expired and has not been changed, and whether the user is locked after three failed logins.

- Password complexity verification

When a user changes the password on the password modification page, the system will verify the password complexity, requiring a combination of uppercase and lowercase letters, symbols and numbers, which should be no less than 8 characters and should not as the same as the user name.

- Ciphertext transfer

The system uses AES symmetric encryption for encryption and decryption to ensure the confidentiality of data transfer.

5.3.4 Session management

- The system has a session termination mechanism. When the user does not respond within 30 minutes, the server will automatically end the session;
- The system will automatically end the session when the user logs off or closes the client.
- The system can limit the maximum number of concurrent sessions and prohibit the same user from logging into the system repeatedly.

Appendix I Indicators Definition

➤ Energy report

Name	Description	Statistical level	Time hierarchy	Data sources
Active Power Production	The sum of the positive active power on the AC side of the energy storage unit.	Station/subsystem	Day/Week/Month/Year/Total	Active Power Production of subsystem
Active Power Consumed	The sum of the reverse active power on the AC side of the energy storage unit.	Station/subsystem	Day/Week/Month/Year/Total	Active Power Consumed of subsystem
Reactive Power Production	The sum of the positive reactive power on the AC side of the energy storage unit.	Station/subsystem	Day/Week/Month/Year/Total	Reactive Power Production of subsystem
Reactive Power Consumed	The sum of the reverse reactive power on the AC side of the energy storage unit.	Station/subsystem	Day/Week/Month/Year/Total	Reactive Power Consumed of subsystem
Energy Conversion Efficiency	The ratio of the amount of active discharge to the amount of active charge	Station/subsystem	Day/Week/Month/Year/Total	Active Power and Reactive Power Production of subsystem

➤ Operation report

Name	Description	Statistica I level	Time hierarchy	Data sources
Discharged Energy @PCS	The total discharge of PCS at the energy storage station	Station/subsystem	Day/Week/Month/Year/Total	Active Power Production of PCS
Charged Energy @PCS	The total charging capacity of the PCS of the energy storage station	Station/subsystem	Day/Week/Month/Year/Total	Active Power Consumed of PCS
Service Factor	The ratio of the operating hours of the energy storage power station to the statistical time.	Station/subsystem	Day/Week/Month/Year/Total	Service hours
Service Hours	The operating time of the station/energy storage unit (the sum of charging hours and discharging hours) and the weighted average of the rated power of each energy storage unit	Station/subsystem	Day/Week/Month/Year/Total	Subsystem operating status Subsystem rated power
Charged Hours	The number of hours when the energy storage power station is in the charging state, and the charging hours of the energy storage power station are weighted according to the rated capacity of the energy storage unit.	Station/subsystem	Day/Week/Month/Year/Total	Subsystem operating status
Discharged Hours	The number of hours that the energy storage power station is in the discharge state, and the discharge hours of the energy storage power station are weighted according to the rated	Station/subsystem	Day/Week/Month/Year/Total	Subsystem rated power

	capacity of the energy storage unit.			
Standby Hours	The number of hours that the energy storage power station is in standby mode, and the standby hours of the energy storage power station are weighted according to the rated capacity of the energy storage unit.	Station/subsystem	Day/Week/Month/Year/Total	Subsystem operating status
Charged Times	A charge should meet: the full-field charging power is greater than 5% of the rated power of the station, and the duration is greater than 15 minutes (the duration of a grid dispatching period), and the default 5% is configurable	Station/subsystem	Day/Week/Month/Year/Total	Subsystem/station active power Subsystem/station rated power
Discharged Times	The primary discharge should be satisfied: the full-field discharge power is greater than 5% of the rated power of the station, and the duration is greater than 15 minutes (the length of a grid dispatching period), and the default 5% is configurable	Station/subsystem	Day/Week/Month/Year/Total	Subsystem/station active power Subsystem/station rated power
Depth of discharge	The weighted average charge/discharge depth of the energy storage power station is comprehensively calculated by using the daily charge-discharge state (SOC) extreme difference and the daily discharge as the weight of the extreme on the basis of the total discharge.	Station/subsystem	Day/Week/Month/Year/Total	Subsystem/station SOC Subsystem/station active power production

Utilization Factor	The ratio of the utilization hours of the energy storage power station to the statistical hours.	Station/subsystem	Day/Week/Month/Year/Total	Utilization hours
Utilization Hours	The sum of the actual amount of electricity (including the actual discharge and actual charging) of all energy storage units in the energy storage power station, divided by the rated capacity (kWh) of the station.	Station/subsystem	Day/Week/Month/Year/Total	Active Power Production Active Power Consumed Rated capacity
Charged Utilization Hours	The number of hours obtained by dividing the total charging capacity of all energy storage units in the energy storage plant by the rated capacity (kWh) of the station.	Station/subsystem	Day/Week/Month/Year/Total	Active Power Consumed Rated capacity
Discharge Utilization Hours	The number of hours obtained by dividing the total discharge of all energy storage units in the energy storage power station by the rated capacity (kWh) of the station.	Station/subsystem	Day/Week/Month/Year/Total	Active Power Production Rated capacity

Appendix II Service Ports

Port	Port type	Service Name	Remarks
22	TCP	sshd	Host
123	UDP	ntp	/
3306	TCP	mysql	Univers docker/Host
6500/6504-6510	TCP	USCADA_upgrade_svr(vsftp)	Univers docker/Host
4005	TCP	USCADA_webservice_svr	Univers docker/Host
4006	TCP	USCADA_cgn_webservice_svr	Univers docker
4506	TCP	USCADA_fc_webservice	Univers docker/Host
4508-4509	TCP	Forecast	Univers docker
6870	TCP	USCADA_data_sync_client	Univers docker
6871	TCP	USCADA_data_sync_client_realtime	Univers docker
6872	TCP	USCADA_data_sync_server	Univers docker
6873	TCP	USCADA_data_sync_server_realtime	Univers docker
6888/6886	TCP/UDP	USCADA_msgbus_svr	Univers docker
7777/6872	TCP	USCADA_monitor	Univers docker
8701	TCP	USCADA_tomcat7/scada_web_gateway	Univers docker/Host
8702	TCP	USCADA_tomcat7_2	Univers docker
8292	TCP	USCADA_WFPCCConfigServiceHost	Univers docker
8890	TCP	USCADA_data_repair_service	Univers docker

8892	TCP	USCADA_config_center_service	Univers docker/Host
9100	TCP	USCADA_agc_bat_ctrl/USCADA_agc_ipower	Univers docker
9300	TCP	USCADA_db_login_service	Univers docker/Host
15000	TCP	USCADA_tenmindata_wssvr	Univers docker
15001	TCP	USCADA_alarmoption_wssvr	Univers docker/Host
2401-2408	TCP	USCADA_iec104	Univers docker
3000-3009	TCP	USCADA_iec102_zf	Univers docker
3000-3099	UDP	USCADA_C_Net	Univers docker
3100	UDP	USCADA_ForwardGapMQ	Univers docker
4880	TCP	USCADA_opc ua server	Univers docker
5020-5023	TCP	USCADA_MODBUS	Univers docker
8079	TCP	USCADA_opc xml-da server	Univers docker
20001-20200	TCP	USCADA_AdsProxy	Univers docker
22221/22222	UDP	USCADA_fe_switch	Univers docker
6501-6503	TCP	DMC	Univers docker
177	UDP	Xmanager	/
6000-6010	TCP	Xmanager	/
5901-5910	TCP	VNC	/
443	TCP	nginx https	Univers docker
6379	TCP	redis	Univers docker/Host
2501-2510	TCP	USCADA_NetGapFTP	Univers docker
5560-5569	TCP	USCADA_NetGapFTP	Univers docker/Host
6660-6669	UDP	USCADA_NetGapFTP 代理	Univers docker
6670-6719	TCP	USCADA_BackwardGapAgent-C	Univers docker

61616	TCP	USCADA_MQ	Univers docker/Host
1883	TCP	USCADA_scada_mqtt_svr	Univers docker/Host
8161	TCP	USCADA_MQ_web	Univers docker/Host
15005	TCP	USCADA_alarm_wssvr	Univers docker
6511	TCP	USCADA_dataquality	Univers docker
6512	TCP	USCADA_dataquality	Univers docker
6513	TCP	USCADA_dataquality	Univers docker
6514	TCP	USCADA_cfg_cntr_ssl_svr	Univers docker
6515	TCP	USCADA_cfg_load_svr	Univers docker
6516	TCP	USCADA_ukey-plugin.jar	Univers docker
6517	TCP	USCADA_ukey-auth.jar	Univers docker
6518	TCP	USCADA_dataset_svr	Univers docker/Host
6519	TCP	USCADA_noise_ctrl_svr	Univers docker
6520	TCP	USCADA_Analysis_Forecast _Module	Univers docker
6521	TCP	/	/
6526	TCP	USCADA_IT_monitor-agent	Univers docker
6527	TCP	USCADA_IT_monitor- transfer	Univers docker
6528	TCP	USCADA_IT_monitor-web.jar	Univers docker
6522	TCP	USCADA_data_cascade_pro ducer	Univers docker
6524	TCP	USCADA_dataset_center_sv r	Univers docker
6523	TCP	USCADA_record_history_svr	Univers docker
6525	TCP	USCADA_auto-check-client	Univers docker
6529	TCP	USCADA_proManSvr	Univers docker

6530	TCP	USCADA_secdata_query_sv r	Univers docker
6531	TCP	emc	Univers docker
6532	TCP	USCADA_tensorflow- detection	Univers docker
6533	TCP	USCADA_db_config_proxy_ svr	Univers docker
6534	TCP	sshd	Host
6535	TCP	USCADA_fdb_remote_servic e	Univers docker/Host
6536	TCP	USCADA_svc_fp_proxy	Univers docker
6537	TCP	USCADA_sc_service	Univers docker
6538	TCP	model_service	Univers docker/Host
6539	TCP	scada_web_service	Univers docker/Host
6540	TCP	update_proxy_svr	Univers docker/Host
6541	TCP	wave_task_mgr(PMS)	Univers docker
6542	TCP	wave_task_mgr(FR)	Univers docker
6543	TCP	wave_task_mgr(LOS)	Univers docker
6544	TCP	tomcat7	Univers docker
6545	TCP	Alarm	Univers docker
6546	TCP	Alarm	Univers docker
6547	TCP	New custom reports	
6544-6549	TCP	Reserved	Univers docker
6550-6552	TCP	PMS	Univers docker
6553	TCP	USCADA_IMA	/
6554	TCP	emc-web	/
6555	TCP	emc-novnc	/

6556	TCP	emc-novnc	/
5500	UDP	emc-center	/
6600	TCP	emc-agent	/
7766	TCP	emc-center	/
6557-6599	TCP	Reserved Host	
6601-6659		Reserved	
8088	TCP	FC/PMS	Univers docker
8090	TCP	FC	Univers docker
7086	TCP	USCADA_influxdb	influxdb docker
7088	TCP	USCADA_influxdb(app docker)	influxdb docker
7089	UDP	USCADA_influxdb(app docker)	influxdb docker
5506	TCP	USCADA_GSP	Univers docker
5507	TCP	USCADA_GSP	Univers docker/Host
5508	TCP	USCADA_GSP	Univers docker
5509	TCP	USCADA_GSP	Univers docker
Active Power Consumed	The sum of the reverse active power on the AC side of the energy storage unit.	Station/subsystem	Day/Week/Month/Year/To tal
Reactive Power Production	The sum of the positive reactive power on the AC side of	Station/subsystem	Day/Week/Month/Year/To tal

	the energy storage unit.		
Reactive Power Consumed	The sum of the reverse reactive power on the AC side of the energy storage unit.	Station/subsystem	Day/Week/Month/Year/Total
Energy Conversion Efficiency	The ratio of the amount of active discharge to the amount of active charge	Station/subsystem	Day/Week/Month/Year/Total

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